

Review

Advances in Perception, Autonomous Operation, and Collaborative Systems for Smart Orchard Robots

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Abstract

Orchard production involves intensive labor requirements, limited operational periods, and highly dynamic and complex working environments. Consequently, the development of intelligent orchard robots has become an important approach to enhancing production efficiency and reducing reliance on manual operations. This review focuses on the demands of autonomous robotic systems operating in challenging orchard scenarios and provides a comprehensive overview of key technologies, including environmental perception and semantic cognition, autonomous navigation and environmental modeling, intelligent task execution, and collaborative robotic systems. Recent advances in fruit and blossom detection, branch and canopy structure perception, multi-modal sensor fusion for localization, semantic mapping, robotic harvesting control, variable-rate spraying, precision pollination, and autonomous intra-row weed management are systematically discussed. Furthermore, emerging technologies such as multi-robot coordination, robot–UAV cooperation, large language models (LLMs), and vision-language models (VLMs) for enhancing decision-making capabilities in agricultural robotics are reviewed. Finally, the existing challenges of orchard robots in terms of perception reliability, long-term autonomous navigation, operational robustness, system-level integration, and standardized performance evaluation are analyzed, followed by discussions on potential future research directions.

Keywords: intelligent orchard robots; environmental sensing; autonomous navigation; precision agricultural operations; multi-robot collaboration; intelligent agriculture

1. Introduction

Orchard production is one of the most labor-intensive components of agricultural production systems and involves diverse operations, including blossom pollination, pest and disease control, weeding, inspection, harvesting, and transportation. Compared with field-crop production, orchard operations are highly seasonal, have narrow operational windows, involve spatially dispersed targets, and remain heavily dependent on manual labor. In recent years, rising labor costs and a declining agricultural workforce have placed substantial pressure on conventional orchard-management practices. In particular, reliance on manual labor alone may be increasingly insufficient to meet the requirements of large-scale, refined, and highly efficient production in harvesting, spraying, pollination, and weeding. Accordingly, the use of robots, sensors, autonomous navigation, machine vision, and intelligent optimization algorithms to increase orchard automation has become a major direction in the development of smart orchards [1–3].



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With the development of smart and digital agriculture, agricultural robots are evolving from single-purpose mechanized equipment into intelligent systems capable of environmental perception, autonomous mobility, precision execution, and task-level decision-making. Previous studies have indicated that autonomous driving systems, intelligent agricultural robots, and multi-robot systems are increasingly being applied to conventionally manual tasks such as seeding, spraying, and harvesting, while agricultural production is progressively moving toward unmanned and intelligent operation [4]. In orchard environments in particular, robotic technologies have been used for fruit harvesting, flower pollination, variable-rate spraying, intra-row weeding, environmental monitoring, and yield estimation, thereby offering technical pathways to alleviate labor shortages, improve operational efficiency, and reduce resource consumption [5,6].

Existing studies have demonstrated a broad range of orchard-robot applications. Harvesting robots mainly address fruit recognition, three-dimensional localization, manipulator path planning, and end-effector control [7]. Spraying robots emphasize target recognition, variable-rate pesticide application, and improved utilization of spray liquid. Pollination robots focus on flower detection, identification of suitable pollination windows, and precise pollen delivery. Weeding robots primarily address weed recognition between rows or individual trees, path tracking, and operational safety [8,9]. For example, research on spraying robots for jujube orchards has noted the large scale of China's jujube industry and the important role of high-density dwarf jujube orchards in Xinjiang. Mid-season management in these orchards still depends heavily on manual labor and is affected by pronounced seasonal and structural labor shortages, creating a clear practical demand for automated orchard robots [10].

At the same time, the capabilities of a single robot the capabilities of a single robot are no longer sufficient to meet the production requirements of large orchards. In extensive orchards, densely distributed task locations, or geographically distributed orchard systems, a single robot often suffers from low operational efficiency, limited coverage, and long task-completion times. Multi-robot collaboration, robot-UAV collaboration, and cross-orchard task scheduling have therefore emerged as new research directions in smart orchards [11,12]. Related studies have shown that agricultural robots and UAVs can be applied to pollination, harvesting, spraying, and monitoring. Ground robots are well suited to close-range precision operations, whereas UAVs are advantageous for aerial spraying and inspection. Coordinated use of the two platforms can exploit their complementary strengths and improve the efficiency of complex agricultural operations [13–15].

Consequently, research on intelligent orchard robots is no longer confined to a single mechanical structure or a single-purpose machine. Instead, it is increasingly oriented toward an integrated technological system comprising multimodal perception, autonomous navigation, precision operation execution, task scheduling, and intelligent decision-making. Achieving robust perception, reliable mobility, accurate execution, and efficient collaboration in complex orchard environments has become a key requirement for advancing orchard robots from experimental validation to practical deployment [16–18].

Orchard environments combine semi-structured and unstructured characteristics. Fruit trees are generally planted in rows and therefore exhibit a degree of spatial regularity; however, canopy morphology, foliage density, fruit distribution, terrain undulation, weed cover, and working-lane width remain highly variable. These environmental characteristics create substantial challenges for orchard robots in perception, localization, navigation, and operation execution [19]. In contrast to indoor or standardized industrial environments, orchard robots must not only identify operation targets but also understand tree structure, traversable areas, obstacle locations, and the spatial relationships between operation targets and robotic actuators.

First, complex orchard environments impose major challenges on perception systems. Fruits, flowers, branches, trunks, leaves, and weeds vary considerably in color, morphology, and spatial distribution and are frequently affected by occlusion, overlap, illumination variation, and background interference. For harvesting robots, fruit-localization errors directly affect grasping success and fruit-damage rates. For pollination robots, the accuracy of flower recognition and pollination-position control influences pollination performance. For weeding and spraying robots, accurate recognition of trunks, weeds, canopies, and target plants is essential for operational safety and resource-use efficiency [20,21]. Studies on semantic mapping have shown that conventional simultaneous localization and mapping (SLAM) can provide relatively accurate geometric maps, but geometric information alone is insufficient to support autonomous orchard operations. Robots must also understand semantic information, such as trees, terrain, and traversable regions, within the map. Second, orchard localization and navigation are challenged by unstable global navigation satellite system (GNSS) signals, sensor degradation, and dynamic environmental changes [22]. GNSS performs well in open farmland, but canopy occlusion, multipath interference, and terrain shielding in orchards can reduce positioning accuracy or cause signal loss. Inertial navigation can provide continuous short-term pose estimates but accumulates drift. Vision-based navigation provides rich environmental information but is vulnerable to illumination, weather, and motion blur. Two-dimensional LiDAR is relatively insensitive to lighting but captures only planar information and may provide insufficient geometry in complex canopies and undulating terrain [23]. To address these issues, current studies increasingly employ multi-sensor fusion involving LiDAR, RGB-D cameras, inertial measurement units (IMUs), wheel odometry, and adaptive Monte Carlo localization (AMCL) to improve localization stability and path-tracking performance in GNSS-constrained orchard environments.

Orchard operations also impose stringent requirements on navigation and execution accuracy. Spraying, pollination, weeding, and harvesting cannot be completed merely by reaching a target area; rather, robots must operate stably along specified paths and at prescribed distances and attitudes [24]. For example, a spraying robot in a high-density jujube orchard must follow predefined operation paths with high-precision cruising and point-specific stopping. Studies have implemented spray-route navigation using LiDAR SLAM, IMU-based inertial navigation, extended Kalman filtering (EKF), AMCL localization, A* path planning, and dynamic window approach (DWA)-based local obstacle avoidance [25]. Field experiments reported navigation-error indices for straight-line travel, point stopping, turning, and S-shaped path tracking, demonstrating explicit requirements for lateral-deviation and heading-angle control in practical orchard operation. Meanwhile, orchard-robot tasks are expanding from single-robot, single-task operation toward multi-robot, multi-task, and multi-scenario systems. Although multi-robot systems can improve operational efficiency, they introduce complex challenges involving task allocation, path conflicts, energy constraints, load balancing, and operation-time limits. Existing studies have emphasized the inherent coupling between task allocation and path planning in agricultural multi-robot operations. If task allocation is considered without accounting for obstacles, terrain constraints, and dynamic interactions among robots, the resulting solution may be difficult to execute effectively in real environments [26]. Joint optimization of task allocation and path planning is therefore an important direction for improving the feasibility and robustness of collaborative agricultural-robot operations.

Collaborative scheduling becomes even more complex in distributed or multi-orchard scenarios. Orchards may be geographically dispersed and differ in area, task density, and environmental conditions. In addition, robot and UAV operations may be subject to precedence constraints. For example, in the task-allocation problem involving pollination robots and spraying UAVs across multiple orchards, robots must complete pollina-

tion before UAV spraying begins, and this precedence requirement markedly increases scheduling complexity [27,28]. The simultaneous involvement of multiple agents, multiple orchards, heterogeneous task densities, and heterogeneous platforms further strengthens the combinatorial-optimization nature of these problems.

Overall, autonomous operation in complex orchard environments faces four principal challenges. First, perceptual targets are complex: fruits, flowers, branches, trunks, and weeds are readily affected by occlusion and environmental variation. Second, the navigation environment is difficult: GNSS constraints, uneven terrain, and canopy occlusion reduce localization stability. Third, operation execution requires high precision: robots must perform accurate, efficient, safe, and minimally damaging operations within confined spaces. Fourth, multi-robot collaborative scheduling is highly complex and must simultaneously account for task duration, path cost, load balance, energy constraints, and precedence relationships. Together, these challenges require orchard-robot research to progress from isolated module optimization toward system-level co-design.

To address the requirements of autonomous agricultural-robot operation in complex orchard environments, this review is organized around the technical chain of environmental perception, autonomous navigation, precision execution, and collaborative decision-making. In contrast to reviews focused on a single robot category or a single operation technology, this article emphasizes the technical interrelationships among the perception, mobility, execution, and decision layers and seeks to capture the overall trend toward systematic development of intelligent orchard robots.

The review first summarizes environmental-perception and semantic-understanding technologies for orchard robots. This section examines methods for recognizing and localizing representative objects, including fruits, flowers, branches, trunks, weeds, and obstacles, with particular emphasis on applications of machine vision, RGB-D sensing, stereo vision, LiDAR, point-cloud processing, instance segmentation, semantic segmentation, and VLMs in complex orchard scenes. Existing studies show that geometric maps alone cannot satisfy the requirements of autonomous operations. Semantic maps provide high-level information on tree layouts, target instances, and traversable regions and thus constitute an important foundation for autonomous navigation and task execution.

Second, this review summarizes localization, navigation, and scene-mapping methods for orchard robots. With emphasis on autonomous mobility in GNSS-constrained environments, this section discusses visual navigation, LiDAR navigation, SLAM-based mapping, multi-sensor fusion localization, semantic map construction, path planning, and path-tracking control. Related studies indicate that multi-sensor fusion can partially compensate for the limitations of individual sensors and improve navigation reliability in complex, occluded, and unstructured orchard scenes.

Third, this review summarizes precision-execution technologies for representative orchard operations, including harvesting, spraying, pollination, and weeding. Key issues include operation-target localization, manipulator control, variable-rate spraying, liquid pollination, intra-row weeding, wind compensation, error calibration, and system evaluation. Comparison across tasks shows that precision execution depends not only on actuator performance but also on close coordination among front-end perception accuracy, navigation stability, and operation-control strategies. Finally, multi-robot collaborative scheduling and intelligent decision-making are reviewed, with emphasis on multi-robot task allocation, joint task-path optimization, multi-trip harvesting schedules, robot-UAV collaboration, multi-orchard operation management, and the use of LLMs and VLMs in robotic task planning. Existing research suggests that single-robot systems are insufficient for large-scale orchard operations. Although multi-robot systems can alleviate efficiency bottlenecks, they also require effective solutions to task allocation, path planning, load

balancing, and energy constraints. Accordingly, this article aims to provide a clear, comprehensive, and engineering-oriented review framework for intelligent orchard robots and to establish a basis for the subsequent analysis of environmental perception, autonomous navigation, precision operation, and collaborative decision-making. To illustrate the hierarchical structure of the technological system and the relationships among technical pathways and operation applications, Table 1 summarizes the core content, representative methods, supported tasks, and primary functions of each technological layer.

Table 1. Key technology framework and representative application tasks of intelligent orchard robots.

Technology Layer	Core Content	Representative Sensors/Methods	Supported Orchard Operations	Primary Role
Environmental perception and semantic understanding	Recognition of fruits, flowers, branches, trunks, weeds, and obstacles	RGB, RGB-D, stereo vision, LiDAR, instance segmentation, semantic segmentation, VLMs	Harvesting, pollination, spraying, weeding, inspection	Acquire operation targets and environmental semantic information
Autonomous navigation and scene mapping	Localization, SLAM, path planning, inter-row navigation, semantic maps	LiDAR SLAM, IMU, AMCL, A*, DWA, visibility graphs	Spraying, weeding, inspection, transportation	Ensure stable robot mobility within orchards
Precision operation execution	Manipulator control, variable-rate spraying, liquid pollination, intra-row weeding	Manipulators, spraying units, PLCs, wind compensation, lateral mechanisms	Harvesting, spraying, pollination, weeding	Convert perception outputs into specific operation actions
Collaborative scheduling and intelligent decision-making	Multi-robot task allocation, robot-UAV collaboration, behavior-tree planning	Evolutionary algorithms, genetic algorithms, artificial bee colony algorithms, LLMs, VLMs	Multi-orchard management, multi-robot harvesting, collaborative spraying	Improve the efficiency of large-scale orchard operations
System evaluation and application validation	Perception accuracy, navigation error, operation outcomes, resource consumption	Localization error, pesticide-saving rate, fruit-set rate, weeding accuracy, completion time	End-to-end operation systems	Provide evidence for evaluating field applicability

For the purpose of this review, a smart orchard robot is defined as an embodied or physically deployed robotic system designed to operate in orchard environments and capable of implementing at least part of an integrated perception–decision–action loop. Such systems may perform autonomous or semi-autonomous functions related to orchard inspection, navigation, harvesting, spraying, pollination, weeding, transportation, or coordinated task execution.

Ground mobile robots constitute the primary focus of this review. Aerial robots are included only when they directly perform orchard operations or cooperate with ground robots in sensing, task planning, scheduling, or execution. Fixed sensing platforms are not regarded as robots but are considered enabling infrastructure when their outputs are directly integrated into robotic perception, localization, planning, or control. Agricultural machinery is included only when it incorporates orchard-specific autonomous sensing, localization, decision-making, or operation-control capabilities.

Individual sensors, perception algorithms, manipulators, end effectors, and control modules are treated as robotic enabling subsystems rather than independent orchard robots. They are included when they have a demonstrated or clearly defined role within a robotic operation pipeline. Similarly, LLMs, VLMs, and other foundation models are considered only when they directly support robot perception, semantic reasoning, task planning, behavior generation, or control. General-purpose artificial-intelligence applications, conventional mechanized equipment, standalone remote-sensing studies, and broader smart-farming

management systems without a direct connection to orchard-robot operation are outside the principal scope of this review.

To distinguish platform types from functional technologies, the reviewed studies are classified along two dimensions:

- (1) a platform dimension comprising ground robots, aerial robots, heterogeneous ground–aerial systems, and enabling robotic subsystems; and
- (2) a functional dimension comprising environmental perception, localization and navigation, precision operation execution, collaborative decision-making, and system-level evaluation.

To improve the transparency and reproducibility of this review, a structured literature search and study-selection process was conducted. The search was completed on 15 June 2026 using SPIS, covering publications from 2002 to 15 June 2026. Search terms covered orchard environments, robotic platforms, and key functions, including perception, navigation, mapping, harvesting, spraying, pollination, weeding, manipulation, multi-robot systems, robot–UAV cooperation, and vision-language models. Search syntax was adapted for each database, and reference lists of relevant reviews and primary studies were also examined.

A PRISMA-type procedure was used for study selection. The initial search identified 215 records. After duplicate removal, titles, abstracts, and full texts were screened. Studies were included if they investigated orchard robots or enabling technologies with a clear role in a robotic perception–decision–execution pipeline and reported sufficiently detailed or quantitative results. Studies focused solely on conventional machinery, standalone remote sensing, general smart-farming management, or algorithms without a clear robotic application were excluded. Editorials, conference abstracts without full texts, non-peer-reviewed reports, and studies lacking sufficient methodological or experimental information were also excluded. In total, 154 records were excluded and 61 core studies were retained.

The selected studies were classified into four categories: environmental perception and semantic understanding, autonomous navigation and scene mapping, precision operation execution, and collaborative scheduling and intelligent decision-making. Studies covering multiple categories were classified according to their main contribution.

Because of substantial differences in crops, orchard structures, robotic platforms, experimental conditions, and evaluation metrics, a formal meta-analysis was not performed. Instead, the studies were qualitatively assessed in terms of quantitative evidence, technological maturity, field-validation scale, environmental robustness, reproducibility, computational requirements, and application relevance. Unreported information was marked as “NR” rather than inferred.

Tables 2–5 summarize representative studies with sufficient quantitative or contextual information. The tables compare research targets, technical methods, experimental settings, main findings, limitations, contextual factors, and evidence profiles. They are intended to clarify applicability and evidence boundaries rather than provide direct performance rankings.

2. Orchard Environmental Perception and Semantic Understanding

Stable and accurate perception and semantic understanding of complex orchard environments are prerequisites for autonomous robotic operation [29]. Unlike industrial environments, orchard scenes contain intertwined fruits, flowers, branches, trunks, leaves, weeds, and ground surfaces and are frequently characterized by occlusion, overlap, illumination variation, scale differences, and depth noise. Perception systems must not only recognize targets but also provide reliable spatial information for manipulator control, precision spraying, liquid pollination, intra-row weeding, path planning, and obstacle avoidance. Consequently, orchard perception is evolving from conventional two-dimensional image

recognition toward multimodal semantic-understanding systems that integrate RGB images, RGB-D data, stereo vision, LiDAR point clouds, and VLMs [30–32].

2.1. Recognition of Fruits, Flowers, and Operation Targets

Fruits, flowers, and canopies are the most direct operation targets for orchard robots [33,34]. Harvesting robots require three-dimensional fruit positions; pollination robots must identify flowers and determine spray locations; and spraying robots must extract target plants or canopy structures from complex backgrounds. Because orchard scenes are affected by foliage occlusion, fruit overlap, and illumination variation, conventional two-dimensional detection alone is insufficient for execution-layer requirements. The integration of object detection, instance segmentation, and three-dimensional localization has therefore become an important trend [35–37].

To address apple localization in complex orchards, Bhattacharya et al. proposed a three-dimensional localization framework based on detection, segmentation, and clustering [38]. Apples are first detected using a VLM, after which instance segmentation is used to obtain apple pixel regions. The pixels are subsequently projected into three-dimensional space, and density-based spatial clustering of applications with noise (DBSCAN) is applied to filter depth noise [38]. The method was evaluated on an RGB-D image dataset containing 219 apples and including indoor and outdoor scenes as well as real and simulated fruit. The mean absolute localization error was 7.7 mm, representing a 74.3% improvement over the best baseline. The approach was also deployed on a field harvesting robot, where it exhibited favorable real-time performance and robustness to occlusion. The principal contribution of this study is the integration of two-dimensional semantic recognition with three-dimensional geometric localization, enabling perception outputs to be used directly for robotic harvesting. Nevertheless, the sample size remains limited, and generalization across cultivars, seasons, and heavily occluded scenes requires further validation.

Hao et al. developed a gripping-assisted flower-detection and wind-compensated pollination robot for precision liquid pollination in kiwifruit orchards. The system used CLIP-assisted automatic annotation to construct a flower-detection dataset and trained a lightweight YOLO model for real-time recognition. It also integrated liquid-spray control and wind-speed and wind-direction compensation modules [39]. Field trials conducted during 2024–2025 showed that, after wind compensation and a second supplemental spray, the fruit-set rate exceeded 82% in all test areas and reached a maximum of 96.0%. This study demonstrates that flower recognition should not be evaluated solely in terms of detection accuracy but should also be jointly assessed with pollination performance, pollen consumption, and compensation for environmental disturbances. A remaining limitation is that mechanical pollination did not fully match manual pollination for some fruit-quality indicators, indicating the need for coordinated optimization of flower perception, spraying mechanisms, and operation strategies.

For spray-target recognition, Hui Liu et al. developed an intelligent multivariable spraying robot for orchards and nurseries. The robot used three-dimensional LiDAR to acquire point clouds of target plants and an improved DBSCAN algorithm to extract plant information. Spray flow rate, air volume, droplet diameter, and spraying direction were then controlled according to target position, shape, and volume [40]. In field trials with canopy diameters of 0.8–1.5 m, tree heights of 1.5–1.8 m, and inter-tree spacing of 2 m, conventional spraying consumed 180–220 g of liquid per nozzle, whereas multivariable spraying consumed 30–35 g, corresponding to an average pesticide-saving rate of an average spray-liquid reduction of approximately 83%. The study shows that three-dimensional target recognition can be used not only for localization but also to derive spray-control parameters, thereby closing the loop from target recognition to variable execution.

Recognition of fruits, flowers, and operation targets is shifting from isolated detection toward an integrated pipeline of recognition, localization, and execution-parameter generation. Future research should focus on improving the practical usability of perception outputs in real robotic systems rather than solely optimizing offline detection metrics.

2.2. Perception of Branches, Canopies, and Tree Structure

Branches, canopies, and overall tree structure provide essential information for orchard robots to understand their surroundings [41,42]. Branches may occlude fruit and interfere with harvesting paths, trunks can serve as spatial references for navigation and weeding, and canopy structure determines spray coverage and liquid deposition. Perception of tree structure is therefore an important link between environmental understanding and precision operation [42–45].

To address semantic segmentation of apple-tree branches under complex orchard conditions, Zhengyang Liu et al. proposed EDSC-HRAFNNet. Built on HRNet, the model introduced an enhanced dynamic snake-convolution module to capture branch bifurcations and elongated topological features, together with HeteroFPN and Parallel-M4 decoding modules to strengthen multiscale feature representation [46]. The authors constructed a multi-scene in situ apple-tree dataset and evaluated the method under eight complex orchard conditions. Precision, recall, Dice, intersection over union (IoU), mean IoU (mIoU), and mean pixel accuracy (MPA) reached 91.50%, 91.71%, 91.60%, 84.51%, 91.72%, and 95.58%, respectively. Relative to HRNet, these metrics increased by 6.98%, 10.09%, 9.21%, 13.50%, 7.25%, and 5.24%, respectively. The method specifically addressed the discontinuous segmentation of slender branches and bifurcated structures and is therefore relevant to obstacle avoidance in harvesting and to robotic pruning. However, its frame rate of 25.1 frames per second (FPS), although sufficient for low-speed apple-harvesting robots, imposes a greater computational burden than lightweight models. Performance also declined in scenes containing yellow apples and under adverse weather conditions, indicating that complex illumination and background similarity continue to affect branch-recognition stability.

From the perspective of trunk perception, Sampurno et al. developed an orchard robot for single-tree-row tracking during intra-row weeding. The robot used a three-dimensional camera and a lightweight YOLO instance-segmentation model to detect trunks and generated a path with a fixed lateral offset relative to the tree row. LiDAR was simultaneously used to trigger a laterally extending weeding mechanism [47]. Experiments showed a mean lateral-tracking root mean square error (RMSE) of 0.329 m and a mean heading RMSE of 4.45 degrees, which were adequate for low-speed weeding navigation. This study demonstrates that trunk detection is not merely a perception task but directly determines the robot trajectory and the safety of the operation mechanism.

Perception of branches, canopies, and tree structure must balance semantic accuracy, topological continuity, and real-time performance [48,49]. Existing methods can recognize complex branch and trunk structures to some extent, but robustness across cultivars, seasons, severe occlusion, and adverse weather remains a major research priority [50,51].

2.3. Detection of Weeds, Obstacles, and Traversable Areas

Detection of weeds, obstacles, and traversable areas is essential for safe mobility and precision operation of orchard robots [52,53]. Orchard floors commonly contain weeds, fallen branches, stones, irrigation facilities, and uneven terrain. Weeds are operation targets for weeding robots, whereas weeds and other obstacles can compromise driving stability and operational safety for harvesting, spraying, and inspection robots [54–56].

To address uncut weeds remaining between trees during intra-row orchard mowing, Sampurno et al. designed a compact intra-row weeding robot. Machine stereo vision was used to detect trunks and achieve single-row tracking, while LiDAR triggered a lateral mechanism that removed weeds between trees [47]. At a low speed of 0.5 m/s, the lateral weeding mechanism achieved an operation accuracy of 84.04%; when the speed increased to 0.8 m/s, accuracy decreased to 76.85%. The authors attributed this reduction to delays between real-time LiDAR point-cloud processing and actuator motion. These results show that weed and obstacle detection must be assessed not only in terms of recognition accuracy but also with respect to temporal synchronization among perception, decision-making, and execution.

Pan et al. addressed obstacle and traversable-area detection from a navigation perspective through an orchard perception and semantic-mapping method. LiDAR SLAM was used to construct an orchard point-cloud map, a three-dimensional object-detection network (3D-ODN) extracted instance-level semantic information such as fruit trees, and terrain analysis was combined with these results to construct a visibility-graph map for global path planning [57]. Tests in a greenhouse, an unstructured apple orchard, and an outdoor litchi orchard showed that the method could support orchard monitoring and selective harvesting. Further experiments found that the viewpoint feature histogram (VFH) feature achieved an IoU of 0.824 but required 2.13 s for feature extraction. By contrast, the proposed three-channel feature method achieved an mIoU of 0.812, with a feature-extraction time of 0.1681 s and a prediction time of 0.0268 s. Thus, a small reduction in accuracy produced a substantial gain in real-time performance, making the method more suitable for online perception and mapping by mobile robots.

The purpose of traversable-area detection is not simply to segment the ground but to transform tree rows, obstacles, and vehicle-motion constraints into a path representation that a robot can execute. Pan et al. used semantic results to identify tree rows and construct a visibility graph. Compared with the timed elastic band planner (TEB-planner), the method achieved shorter global-planning times and more reasonable paths and could define turning regions for Ackermann-steered vehicles [57]. This study indicates that orchard perception should advance from identifying environmental elements toward generating executable navigation representations.

Detection of weeds, obstacles, and traversable regions is a critical step in the transition from environmental perception to autonomous mobility. Future studies should further improve stability under dense weed cover, muddy ground, occlusion by orchard infrastructure, and dynamic obstacles [58–60].

2.4. Multimodal Fusion and Three-Dimensional Semantic Representation

No single sensor often cannot fully satisfy all perceptual requirements of orchard robots [61]. RGB cameras provide color and texture information but are highly sensitive to illumination. RGB-D and stereo-vision systems provide depth but are constrained by sensing range and noise. LiDAR is robust to illumination changes and is suitable for mapping and structural extraction, but it lacks texture semantics. Multimodal fusion has therefore become an important direction for orchard-environment understanding [62–64].

The semantic-mapping method of Pan et al. illustrates the advantages of integrating LiDAR point clouds with deep learning. The method directly processed the point-cloud map

generated by LiDAR SLAM, used 3D-ODN to extract fruit-tree instances, and combined these results with terrain analysis to generate a semantic map and visibility graph [57]. Compared with approaches that reproject two-dimensional image semantics into three-dimensional space, this method reduced calibration errors and data-transformation costs and was better suited to large-scale orchard mapping. Its main contribution was to elevate point-cloud detections into a three-dimensional semantic representation that could directly support autonomous navigation. Figure 1 illustrates the workflow for orchard semantic mapping based on three-dimensional point clouds.

The multivariable spraying robot developed by Hui Liu et al. further illustrates the integration of three-dimensional perception and execution control. The system used three-dimensional LiDAR to extract target-plant point clouds and converted target volume and spatial position into control variables, including spray flow rate, air volume, droplet diameter, and direction [40]. Experiments showed that multivariable spraying not only reduced liquid consumption but also improved penetration and distribution uniformity through adjustment of air volume and droplet diameter compared with spraying based solely on variable flow rate. This study demonstrates that the value of three-dimensional semantic representation extends beyond target recognition to the optimization of operation quality.

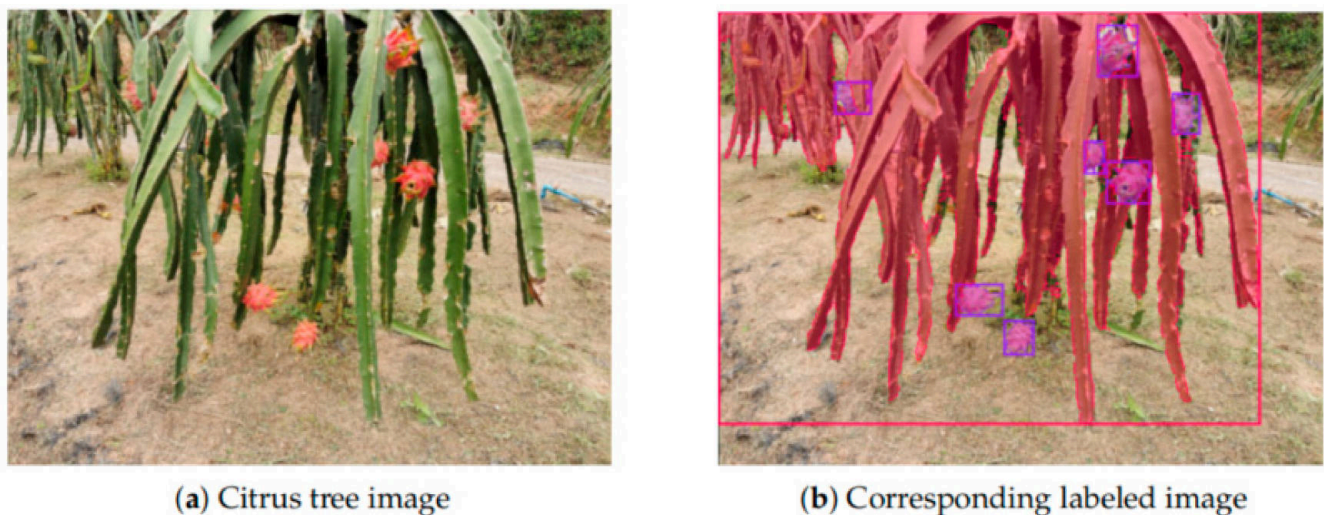


Figure 1. Schematic diagram of semantic annotation [29].

The apple-localization method of Bhattacharya et al. combined VLM-based detection, instance segmentation, RGB-D projection, and DBSCAN clustering to establish a fusion pipeline from two-dimensional semantics to three-dimensional geometry [38]. Sampurno et al. used a three-dimensional camera for trunk detection and path generation and LiDAR for distance estimation and activation of the weeding mechanism, illustrating functional specialization among sensors [47]. Together, these studies show that multimodal fusion is not simply a matter of adding sensors; rather, perceptual responsibilities must be assigned according to task requirements, and temporal synchronization, spatial calibration, and consistency of control responses across modalities must be ensured.

2.5. Foundation-Model-Enabled Understanding of Orchard Scenes

Vision-language models and LLMs provide new technical pathways for orchard-scene understanding [65,66]. Conventional vision models generally depend on large manually annotated datasets, support fixed task categories, and have limited cross-scene transferability. Foundation models offer stronger semantic representation and transfer capabilities and can reduce annotation costs, enable open-vocabulary recognition, and support more flexible task planning when integrated with robotic-control frameworks [67,68].

Bhattacharya et al. used a foundation VLM for apple detection and combined it with instance segmentation and DBSCAN clustering for three-dimensional localization, ultimately achieving a mean absolute localization error of 7.7 mm [38]. This result shows that VLMs are not limited to coarse-grained semantic recognition; when combined with geometric computation, they can also support high-precision localization for harvesting. Hao et al. used CLIP to assist in constructing a kiwifruit-flower detection dataset and then trained a lightweight YOLO model for real-time recognition, thereby reducing manual annotation requirements. Field trials indicate that the value of foundation models in agriculture lies not only in direct target recognition but also in improving the efficiency of data preparation and model development.

Although the AGRI-BT robot proposed by Salman et al. was designed for greenhouse operation, its technical architecture is informative for intelligent orchard robots. The authors constructed 1187 pairs of natural-language agricultural task descriptions and expert-validated XML behavior trees and applied low-rank adaptation (LoRA) to CodeLlama-7B for domain adaptation, enabling the model to convert natural-language tasks into executable behavior trees. CLIP was integrated for visual-semantic judgment [69]. ROS 2 simulations showed 100% XML structural validity, 87.5% semantic correctness, 82.0% task alignment, and an overall weighted validation score of 91.4%. For navigation tasks, specific task descriptions reduced tracking error from 0.208 to 0.135 m, a decrease of 35.1%. This study indicates that foundation models can support not only perception but also task planning and perception-driven decision-making.

Nevertheless, substantial limitations remain in the application of foundation models to agricultural robots [70]. Salman et al. reported that task alignment for complex multistage tasks was lower than both structural validity and semantic correctness, primarily because complex agricultural tasks accounted for only a small proportion of the dataset. CLIP achieved a confidence of 84.1% on natural plant images but only 25–30% on Gazebo-simulated greenhouse images, revealing a domain gap between simulated and real agricultural imagery [71,72]. Foundation models therefore cannot be separated from agriculture-specific data, lightweight deployment, and constraints imposed by robotic execution. A more practical future direction is to combine the semantic generalization capabilities of foundation models with specialized detectors, three-dimensional point-cloud processing, semantic maps, and behavior-tree control to establish orchard-scene understanding systems that are verifiable, deployable, and closed-loop.

In summary, orchard perception has developed along multiple technical pathways, ranging from object detection and structural segmentation to semantic mapping and foundation-model enhancement. These approaches differ in perception targets, technical solutions, and application scenarios. Table 2 summarizes representative studies to systematically compare their performance characteristics and applicability boundaries.

Table 2. Comparison of representative studies on orchard environmental perception and semantic understanding.

Research Target	Representative Study	Perception Method	Data/Experimental Setting	Main Result	Evaluation and Limitations	Reported Contextual Variables and Gaps	Standardized Evidence Profile (Maturity; Field Scale; Robustness; Reproducibility; Computation; Applicability; NR = Not Reported)
Three-dimensional apple localization	Bhattacharya et al.	VLM detection, instance segmentation, RGB-D projection, DBSCAN clustering	219 apples covering indoor and outdoor scenes and real and simulated samples	Mean absolute localization error of 7.7 mm; 74.3% improvement over the best baseline	Suitable for harvesting-manipulator localization, but limited sample size and insufficient validation across cultivars and under severe occlusion	Apple; 219 targets; indoor and outdoor; real and simulated samples; field-robot deployment also reported. Illumination was mixed but not standardized; quantitative occlusion level, travel speed, terrain complexity, and validation duration: NR.	Field prototype; 219 apples plus robot deployment; mixed scenes, but severe occlusion and cultivar transfer NR; method/metrics reported, data/code NR; real-time, hardware/FPS NR; harvesting-specific.
Kiwifruit-flower recognition	Hao et al.	CLIP-assisted annotation and lightweight YOLO detection	Field trials in kiwifruit orchards during 2024–2025	Fruit-set rate above 82% and up to 96.0% after wind compensation and supplemental pollination	Recognition was evaluated together with pollination performance, but mechanical-pollination efficiency still requires improvement	Kiwifruit; open-orchard field trials during 2024–2025, supported by wind-tunnel calibration. Dataset size, illumination category, quantitative occlusion level, robot travel speed, terrain complexity, and continuous validation duration: NR.	Field prototype; orchard trials in 2024–2025 plus wind-tunnel tests; wind robustness tested, lighting/occlusion/terrain NR; data/code/hardware NR; lightweight model; kiwifruit-pollination specific.
Target-plant recognition	Hui Liu et al.	Three-dimensional LiDAR and improved DBSCAN point-cloud clustering	Spraying trials in orchards and nurseries	30–35 g of liquid for multivariable spraying versus 180–220 g for conventional spraying; 83% pesticide saving	Point-cloud recognition directly generated spraying parameters, but canopy-volume estimation remained sensitive to tree architecture	Orchard and nursery field spraying; canopy diameter 0.8–1.5 m, tree height 1.5–1.8 m, and inter-tree spacing 2 m. Crop/cultivar, dataset size, illumination, occlusion level, travel speed, terrain complexity, and validation duration: NR.	Field prototype; orchard/nursery trials with canopy geometry reported; cultivar/weather/terrain NR; control variables/results reported, data/code/hardware NR; processing rate NR; comparable-canopy spraying.
Apple-branch segmentation	Zhengyang Liu et al.	EDSC-HRAFNNet semantic segmentation	Eight complex apple-orchard scenarios	Precision 91.50%, recall 91.71%, IoU 84.51%, and mIoU 91.72%	Suitable for branch-structure understanding, but performance declined for yellow-apple scenes and adverse weather	Apple; in-situ multi-scene dataset covering eight complex orchard conditions; yellow-apple scenes and adverse weather were included. Dataset size, standardized illumination/occlusion levels, robot speed, terrain complexity, and validation duration: NR.	Validated perception model; eight orchard conditions; adverse weather and yellow apples tested with performance loss; data/code NR; 25.1 FPS with relatively high load; low-speed branch perception.
Trunk recognition and row tracking	Sampurno et al.	Three-dimensional camera and YOLO instance segmentation	Navigation trials for intra-row orchard weeding	Lateral-tracking RMSE of 0.329 m and heading RMSE of 4.45 degrees	Trunk recognition supported navigation, but execution accuracy decreased at higher speed	Real-orchard intra-row weeding and row tracking; low-speed operation, with operation results also reported at 0.5 and 0.8 m/s. Crop/cultivar, dataset size, illumination, quantitative occlusion level, terrain complexity, and validation duration: NR.	Field prototype; real-orchard tests at 0.5 and 0.8 m/s; GNSS-independent, broader crop/terrain transfer NR; data/code NR; delay limits speed; clearly defined-row weeding.
Semantic mapping	Pan et al.	LiDAR SLAM, 3D-ODN, terrain analysis, and visibility graph	Greenhouse, apple-orchard, and litchi-orchard environments	Three-channel feature mIoU of 0.812; feature extraction in 0.1681 s and prediction in 0.0268 s	Converted perception results into a navigation graph, but parameters require adaptive adjustment for different orchards	Greenhouse, unstructured apple orchard, and outdoor litchi orchard; terrain analysis and a 10 m × 10 m local map with a 128 × 128 feature map were reported. Travel speed, standardized illumination/occlusion levels, and field-validation duration: NR.	Multi-environment prototype; greenhouse, apple, and litchi tests; standardized lighting/occlusion and duration NR; code/data NR; 0.195 s total reported processing; orchard-specific tuning required.

3. Autonomous Navigation and Scene Mapping for Orchard Robots

Autonomous navigation and scene mapping provide the basis for orchard robots to progress from isolated point operations to continuous field operations. Orchard robots can perform spraying, weeding, harvesting, inspection, and transportation only when they can reliably estimate their own pose, understand orchard structure, and generate executable paths. Compared with open farmland, orchards are characterized by canopy occlusion, narrow inter-row spaces, irregular tree arrangements, uneven terrain, foliage obstruction, and unstable GNSS signals. Navigation systems must therefore achieve not only localization and obstacle avoidance but also task-specific control of path, attitude, and working distance. Accordingly, orchard navigation research has evolved from single-sensor localization toward autonomous mobility systems jointly supported by vision, LiDAR, IMUs, wheel odometry, and semantic maps [73–77].

3.1. Environmental Characteristics and Technical Requirements of Orchard Navigation

Orchard environments are generally semi-structured: tree rows exhibit a degree of regularity, whereas canopies, branches, ground surfaces, and obstacles remain highly uncertain. When operating in orchards, robots must maintain a stable heading within narrow rows and switch paths between trees, at row ends, or within turning areas [78,79]. For spraying robots, navigation errors alter the distance between nozzles and the canopy and consequently affect spray deposition. For weeding robots, excessive lateral deviation may result in missed weeds or tree damage. For harvesting and inspection robots, path stability directly influences the manipulator workspace and the consistency of data acquisition [80–82].

Research on spraying robots for high-density jujube orchards clearly illustrates these requirements [83,84]. For predefined route cruising in densely planted jujube orchards, researchers developed an autonomous navigation system based on LiDAR SLAM and IMU-based inertial navigation and fused IMU and LiDAR localization information using an EKF [85]. LiDAR was used for orchard perception and SLAM mapping, AMCL localized the robot within the map, and A* and DWA algorithms were used to generate and track spraying paths. Field experiments showed that, during travel along a 15 m straight path, the mean deviation along the X-axis was 3.41 cm and the mean heading angle was 1.25 degrees. Within a 1.5 m point-stopping region, the mean X-direction deviation was 1.98 cm and the mean heading angle was 0.77 degrees. During a turn with a radius of 2 m, the mean X-axis deviation was 1.17 cm, the parallel-trajectory distance deviation was 4.79 cm, and the mean heading angle was 3.31 degrees. Along an S-shaped path, the mean X-axis deviation was 1.8 cm and the mean heading angle was 1.6 degrees [86–88]. These results demonstrate that orchard navigation systems must simultaneously satisfy the requirements of straight-line cruising, point stopping, turning, and complex path tracking; arrival at a target point alone is not an adequate evaluation criterion [89,90].

Weeding scenarios impose additional navigation requirements [91]. Intra-row weeding robots must maintain a fixed lateral offset from a single tree row and control laterally moving actuators when approaching trunks [92–94]. The single-tree-row-tracking robot developed by Sampurno et al. used a three-dimensional camera and a lightweight YOLOv8 instance-segmentation model to detect trunks, extract a forward navigation path, and maintain a lateral offset of 0.8 m from the tree row [47]. Simulation and field experiments yielded a navigation RMSE of 0.329 m and demonstrated intra-row weeding navigation without GNSS. This study shows that orchard navigation is not solely a global-localization problem but is also closely related to control of the relative position between the robot and its operation mechanism.

The technical requirements of orchard navigation systems can be summarized as follows: stable localization in GNSS-constrained environments; reliable path extraction

from semi-structured tree rows; turning and obstacle avoidance within confined spaces; and maintenance of task-appropriate lateral distance, heading angle, and operating attitude. Figure 2 illustrates inter-row cruising and point-specific operation of an orchard robot in a high-density orchard.

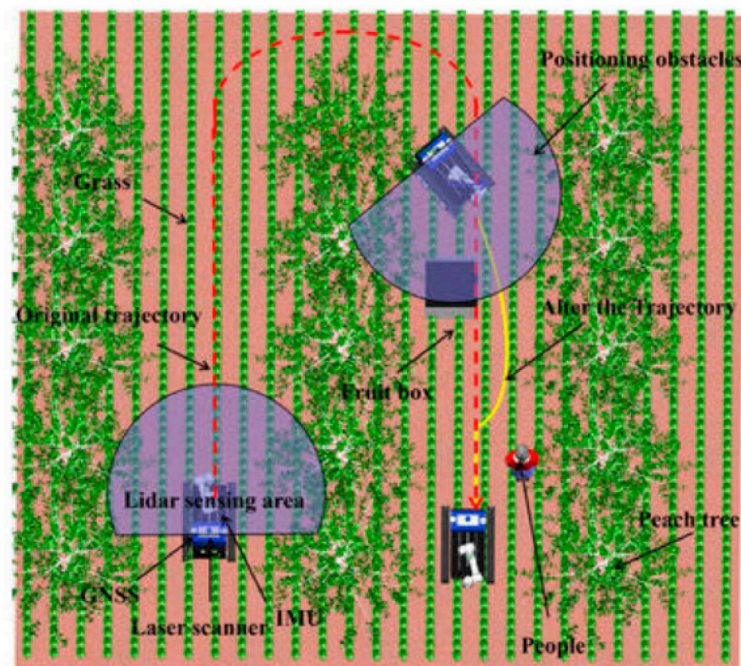


Figure 2. Schematic Diagram of Laser SLAM Navigation and Obstacle Avoidance System [93].

3.2. Vision and LiDAR-Based Navigation Methods

Vision and LiDAR are the two most widely used sensing modalities for orchard-robot navigation [95]. Vision sensors provide rich color, texture, and semantic information and are therefore suitable for trunk recognition, tree-row extraction, and object detection. LiDAR provides stable range measurements and geometric-structure information, is insensitive to illumination changes, and is well suited to mapping, obstacle avoidance, and extraction of inter-row structure. Each modality offers distinct advantages and limitations [96,97].

For vision-based navigation, Sampurno et al. combined a three-dimensional camera with a lightweight YOLOv8 instance-segmentation model for trunk detection and single-row tracking [47]. The method identified trunks on one side, extracted the tree row ahead of the robot, and generated a trajectory with a fixed lateral offset. The study also compared the performance of visual-detection models on an edge device. The lightweight YOLOv8p-seg model achieved a mean average precision (mAP) of 87.6% and 39.07 FPS for trunk detection, satisfying real-time navigation requirements. End-to-end latency from visual perception to steering control was approximately 27.64–30.64 ms, demonstrating the feasibility of vision-based navigation for low-speed orchard operations. The principal advantage is that semantic trunk recognition is directly converted into a navigation-control input, making the method suitable for intra-row operation where GNSS is unavailable and tree-row structure is evident. However, the visual system remains sensitive to illumination, occlusion, and variations in trunk appearance, and actuator accuracy decreases at higher travel speeds.

LiDAR navigation exhibited high stability in the jujube-orchard spraying robot. LiDAR was used for environmental perception, SLAM mapping, extraction of inter-row operation points, and localization assistance [98]. Inter-row points detected by LiDAR were clustered and fitted to extract operation paths; A* performed global path search, while DWA handled local path tracking and obstacle avoidance. In field trials, the robot maintained centimeter-level lateral deviations on straight, point-stopping, turning, and S-shaped paths, demonstrating the

advantages of LiDAR for high-precision cruising in densely planted orchards. A limitation is that two-dimensional or low-channel-count LiDAR provides insufficient representation of canopies, uneven ground, and three-dimensional obstacles. Future systems should therefore incorporate richer three-dimensional point-cloud information [99,100].

In general, vision-based navigation is better suited to extracting semantic targets and tree-row features, whereas LiDAR is more suitable for ranging, mapping, and obstacle avoidance. No single approach can cover all real orchard conditions. Vision offers a cost advantage under stable lighting and when trunks have distinctive features; LiDAR is more robust when illumination varies substantially, environmental structure is complex, or high-precision maps are required. Future studies should emphasize complementary fusion of visual semantics and LiDAR geometry.

3.3. Multi-Sensor Fusion Localization and SLAM

Canopy and terrain occlusion reduce GNSS reliability in orchards, making multi-sensor fusion localization and SLAM important directions for autonomous navigation [101,102]. Vision alone is vulnerable to illumination changes, LiDAR alone may degrade in areas with limited geometric features, and inertial navigation alone accumulates drift [103]. Fusion of IMUs, wheel odometry, RGB-D sensing, LiDAR, and AMCL can improve localization continuity and mapping consistency [104–106].

The jujube-orchard spraying-navigation system combined LiDAR SLAM, IMU-based inertial navigation, EKF, and AMCL localization [107–109]. The IMU provided short-term attitude and motion information, LiDAR supplied environmental geometric constraints, the EKF fused localization information, and AMCL performed global localization in a previously constructed map. Field results showed that the system met the cruising-accuracy requirements of spraying in densely planted jujube orchards [110,111]. Its significance lies in the integration of multi-source localization, path planning, and operation-path tracking within a complete spraying-robot system rather than in validation of an isolated localization algorithm.

Chang et al. proposed a low-cost multi-sensor fusion SLAM navigation system for an autonomous orchard-weeding robot [60]. The system fused RGB-D point-cloud information with two-dimensional LiDAR to construct a two-dimensional occupancy grid and tightly coupled visual-inertial odometry with wheel odometry to mitigate feature degradation in unstructured environments. AMCL was introduced to improve global-localization stability, and Cartographer enhanced mapping accuracy and long-term consistency. In real-orchard experiments, the mean absolute pose error was 0.486 m with a standard deviation of 0.328 m. At travel speeds of 0.5–1.5 m/s, the mean heading deviation remained below 7 degrees, while longitudinal and lateral localization errors were limited to 8 and 7 cm, respectively. These results show that multi-sensor fusion can provide usable navigation in GNSS-denied orchards and is particularly suitable for low-cost weeding robots. However, the absolute pose error remained close to 0.5 m; high-precision harvesting or near-tree operations therefore require additional correction through local visual servoing or close-range perception.

Pan et al. used LiDAR and an IMU with LIO-SAM to construct an orchard point-cloud map and deliberately excluded GPS during mapping because orchard obstacles reduce GPS stability and accuracy [57]. A mobile robot first collected LiDAR data and generated the point-cloud map; fruit-tree detection, terrain analysis, and semantic processing were then performed on the map to create a high-level semantic representation for navigation. In contrast to SLAM systems that output only geometric occupancy grids, this approach further transformed the SLAM result into a scene structure suitable for autonomous operation.

Multi-sensor fusion localization and SLAM improve navigation continuity and resistance to disturbance in orchards but also increase the burden of sensor calibration, temporal

synchronization, computation, and system complexity [112–114]. Future system design should therefore select an appropriate fusion level according to task requirements. General inspection and weeding may use low-cost fusion of RGB-D sensing, two-dimensional LiDAR, and wheel odometry, whereas high-precision harvesting and complex scene mapping are better served by three-dimensional LiDAR and semantic SLAM.

3.4. Semantic Map Construction and Path Planning

Conventional SLAM primarily provides geometric maps, whereas orchard robots need to know not only where obstacles are located but also where tree rows and traversable areas are situated, how each tree can be accessed, and in which direction operations should be performed [115–117]. Semantic maps therefore serve as an important intermediate layer linking perception, navigation, and operation tasks.

The orchard-perception and semantic-mapping framework proposed by Pan et al. is representative [57]. The framework comprises mapping, perception, and semantic-processing stages. First, LiDAR and LIO-SAM are used to construct an orchard point-cloud map. Second, 3D-ODN detects and localizes fruit trees within the map. Finally, cloth simulation filtering (CSF) extracts terrain information, and the fruit-tree detections are used to classify the point cloud into fruit trees, traversable terrain, and other obstacles. A semantic map containing tree distribution, row structure, and access nodes is then constructed. Because the method directly processes a three-dimensional point-cloud map, it avoids reprojection of two-dimensional image semantics into three-dimensional space and reduces the errors and computational overhead associated with additional calibration and data conversion [118,119].

For path planning, the study used the Hough transform to detect tree columns, determined inter-row corridors from their positions, and constructed a visibility graph as a high-level navigation graph [120]. Because vehicles cannot easily turn within narrow rows, one-way travel constraints were assigned within the corridors, and access points with opposite orientations were generated on both sides of each tree column, enabling the robot to form feasible routes around the rows. Experiments were performed in a greenhouse, an apple orchard, and a litchi orchard [121]. With a 10 m × 10 m local map and a 128 × 128 feature map, the method achieved an mIoU of 0.812, a feature-extraction time of 0.1681 s, and a prediction time of 0.0268 s. Compared with the 2.13 s processing time of the VFH feature, the proposed approach substantially improved efficiency while retaining good accuracy.

The semantic map was also compared with TEB-planner for path planning. In the apple orchard, the visibility graph required an average of 4 ms, compared with 34 ms for TEB-planner; the corresponding times were 6 and 45 ms in the greenhouse and 3 and 31 ms in the litchi orchard [122,123]. In addition to faster planning, the visibility graph accounted for the vehicle orientation at the target and produced paths better aligned with orchard-operation directions. These results show that path planning for orchard operations should not simply apply a general-purpose local planner; tree-row structure, vehicle-motion direction, and the sequence of operation visits should be incorporated into the map representation. To enhance the robustness and adaptability of the improved algorithm in the complex environment of the orchard, after the initial path is generated, the path quality is further improved through multiple optimizations. As shown in Figure 3, this is the graph of the path's three optimizations.

In addition to localization and geometric obstacle avoidance, terrain-aware navigation is becoming increasingly important for orchard robots operating on sloped, uneven, muddy, or vegetation-covered ground. Recent approaches use three-dimensional LiDAR, stereo vision, or RGB-D sensing to construct local elevation and traversability maps. Terrain properties such as slope, surface roughness, step height, ground clearance, and disconti-

nities can then be converted into navigation costs, allowing the robot to avoid regions that may cause rollover, chassis collision, excessive vibration, or loss of traction. Semantic information can further distinguish apparently similar geometric surfaces, such as firm soil, loose soil, dense grass, mud, irrigation channels, and fallen branches, which may have substantially different effects on vehicle mobility.

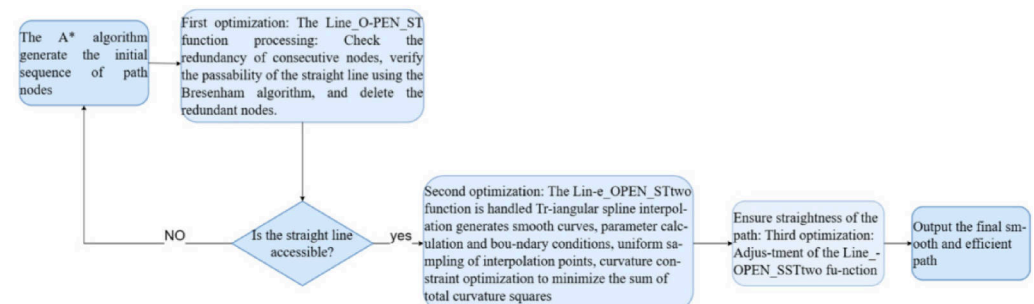


Figure 3. Three Optimization Diagrams of the Path [93].

Terrain assessment based solely on external perception may nevertheless be unreliable when vegetation conceals the ground or when soil conditions change after rainfall. Therefore, newer approaches increasingly combine exteroceptive sensing with proprioceptive information from IMUs, wheel encoders, motor current, suspension displacement, or contact and force sensors. Disagreement between commanded and measured motion can be used to estimate wheel slip, sinkage, or reduced traction. This information can update the traversability map online and support adaptive velocity control, traction control, and local replanning. Such perception–mobility feedback is particularly relevant to orchard spraying and harvesting platforms, whose stability may be affected by high centers of gravity, liquid movement, or changing payloads.

Risk-aware planning and control provide another direction for mobility in challenging environments. Instead of selecting the geometrically shortest route, terrain-aware planners incorporate slope, roughness, predicted slip, energy consumption, vehicle stability, and uncertainty into the path cost. Model predictive control and adaptive path-tracking methods can then adjust travel speed, steering commands, and vehicle attitude according to local terrain conditions. At the platform level, tracked mechanisms, independently driven wheels, active or articulated suspension, and adjustable chassis structures may improve mobility over uneven ground, although they also increase mechanical complexity, power consumption, and maintenance requirements. Future studies should evaluate these approaches through long-duration trials across different soil moisture conditions, slopes, seasons, and orchard structures and should report not only localization and tracking errors but also slip ratio, vibration, energy use, intervention frequency, stability margins, and mission-completion reliability.

Semantic map construction converts a complex orchard into a structured representation that robots can understand, plan within, and execute. A limitation is that parameter settings remain environment dependent. For example, local-map size, feature-map resolution, and ground-segmentation parameters all affect detection accuracy and computational efficiency. Stronger adaptive capabilities are required for different tree species, row spacings, and terrain conditions.

3.5. Robustness, Real-Time Performance, and Generalization of Navigation Systems

Orchard-robot navigation systems must ultimately operate reliably over long periods in real environments; robustness, real-time performance, and generalization are therefore critical evaluation criteria [124]. Robustness refers to the ability to maintain localization and path

tracking under changes in illumination, canopy occlusion, uneven terrain, and sensor noise. Real-time performance requires perception, localization, planning, and control to be updated promptly while the robot is moving. Generalization determines whether a system can transfer from one orchard to different species, tree forms, and cultivation systems [125,126].

Existing experiments indicate that multi-sensor fusion is the primary means of improving robustness. The jujube-orchard spraying robot combined LiDAR, an IMU, EKF, and AMCL to achieve centimeter-level path tracking. The orchard-weeding robot fused RGB-D, two-dimensional LiDAR, visual-inertial odometry, wheel odometry, and AMCL and maintained longitudinal and lateral errors below 10 cm at 0.5–1.5 m/s. The single-tree-row-tracking robot used three-dimensional vision for trunk detection and LiDAR to trigger the operation mechanism, enabling near-tree operation without GNSS. Collectively, these studies show that orchard navigation should not rely on a single information source but should exploit the complementarity of geometric, semantic, and motion information [127–129].

Different studies have adopted different trade-offs to achieve real-time performance. The lightweight YOLOv8p-seg model used by Sampurno et al. achieved 39.07 FPS, with a perception-to-control delay of approximately 27.64–30.64 ms, making it suitable for low-speed weeding navigation [47]. The three-channel point-cloud feature method of Pan et al. required 0.1949 s; although slower than pure vision in frame rate, it provided efficient point-cloud semantic mapping [57]. Visibility-graph path planning required only 3–6 ms across different orchard environments, substantially less than the 31–45 ms required by TEB-planner. These results indicate that real-time optimization depends not only on model lightweighting but also on an appropriate representation of the task.

Generalization remains a weakness of current systems. Although Pan et al. validated their semantic-mapping framework in a greenhouse, an apple orchard, and a litchi orchard, they noted that local-map size and feature-map resolution may require adaptive adjustment for different orchard environments and recommended extending the method to vineyards and field crops. The single-tree-row-tracking method of Sampurno et al. is suitable for weeding where rows are clear and the operation offset is fixed, but visual recognition and actuation timing may become unstable under severe trunk occlusion, irregular rows, or high-speed travel.

Autonomous navigation for orchard robots is progressing from localization and obstacle avoidance toward scene understanding and operation-oriented path planning [130,131]. Future research should strengthen three areas: improving the stability of multi-sensor fusion systems under severe occlusion, sloped terrain, and dynamic obstacles; reducing the computational cost of semantic mapping and path planning to support edge-computing platforms; and establishing cross-orchard, cross-species, and cross-season navigation benchmarks to promote the transition from single-scene experiments to long-term deployment [132,133].

Overall, autonomous orchard-navigation technologies have developed along multiple pathways, including multi-sensor fusion localization and semantic-map-driven planning, with substantial differences in localization accuracy, hardware cost, and applicable scenarios. Table 3 summarizes representative studies to clarify the technical characteristics and performance boundaries of these navigation methods.

Table 3. Comparison of representative methods for autonomous navigation and scene mapping of orchard robots.

Navigation/ Mapping Category	Representative Study	Sensors and Algorithms	Experimental Setting	Key Performance Indicators	Applicability	Crop, Environment, and Operating Context	Standardized Evidence Profile (Maturity; Field Scale; Robustness; Reproducibility; Computation; Applicability; NR = Not Reported)
LiDAR-IMU fusion navigation	Li Yufeng et al.	LiDAR SLAM, IMU, EKF, AMCL, A*, and DWA	Spraying-route navigation in a densely planted jujube orchard	Mean X-axis deviation of 3.41 cm on a 15 m straight path and 1.8 cm on an S-shaped path	Suitable for high-precision cruising and point-specific operation in densely planted orchards	Densely planted jujube orchard; field validation on a 15 m straight route, point stopping, a 2 m-radius turn, and an S-shaped route. Travel speed, illumination, terrain complexity, and continuous validation duration: NR.	Field prototype; one dense jujube orchard with four route types; GNSS constraint addressed, lighting/terrain/duration NR; data/code/hardware NR; real-time resource load NR; dense-orchard spraying navigation.
Single-tree-row- tracking navigation	Sampurno et al.	Three-dimensional camera, YOLOv8 instance segmentation, and LiDAR triggering	Intra-row weeding robot	Navigation RMSE of 0.329 m and heading RMSE of 4.45 degrees	Suitable for low-speed operation in orchards with clearly defined tree rows	Real-orchard intra-row weeding; clearly defined single tree rows; fixed lateral offset of 0.8 m; GNSS-independent operation; operation accuracy reported at 0.5 and 0.8 m/s. Crop/cultivar, illumination, terrain complexity, and validation duration: NR.	Field prototype; real-orchard tests at 0.5 and 0.8 m/s; GNSS-independent in clear rows, broader transfer NR; data/code NR; higher-speed accuracy loss; low-speed near-tree navigation.
Multi-sensor fusion SLAM	Chang et al.	RGB-D, two-dimensional LiDAR, VIO, wheel odometry, AMCL, and Cartographer	Weeding robot in a real orchard	Mean absolute pose error of 0.486 m; longitudinal and lateral errors of 8 and 7 cm	Suitable for low-cost navigation in GNSS-denied environments	Real-orchard weeding in a GNSS-denied/unstructured environment; travel speeds of 0.5–1.5 m/s. Crop type, illumination, quantitative occlusion level, terrain complexity, and validation duration: NR.	Field prototype; real-orchard tests at 0.5–1.5 m/s; GNSS-denied/unstructured conditions tested, crop/lighting/terrain/duration NR; data/code/hardware NR; low-cost stack; weeding navigation.
Point-cloud semantic mapping	Pan et al.	LiDAR, LIO-SAM, 3D-ODN, CSF, and visibility graph	Greenhouse, apple orchard, and litchi orchard	mIoU of 0.812; feature extraction in 0.1681 s and prediction in 0.0268 s	Suitable for extending geometric maps into semantic maps	Greenhouse, apple orchard, and litchi orchard; GPS excluded during mapping; 10 m × 10 m local map and 128 × 128 feature map. Travel speed, illumination/ occlusion levels, terrain complexity, and validation duration: NR.	Multi-environment prototype; greenhouse, apple, and litchi mapping without GPS; standardized lighting/occlusion and long-term stability NR; code/data NR; 0.195 s total processing; semantic mapping.
Semantic path planning	Pan et al.	Tree-row recognition, Hough transform, and visibility-graph path planning	Apple orchard, greenhouse, and litchi orchard	Planning time of 4 ms in the apple orchard versus 34 ms for TEB-planner	Suitable for operation-oriented path planning in structured tree rows	Apple orchard, greenhouse, and litchi orchard; structured tree-row/access-node representation and operation-oriented routes. Travel speed, illumination, terrain complexity, and field-validation duration: NR.	Prototype planner; apple, greenhouse, and litchi tests; structured rows tested, dynamic obstacles/seasonal changes NR; code/data NR; 4 ms versus 34 ms; structured-orchard global planning.

4. Precision Operation Technologies for Orchard Robots

Precision operation execution is the ultimate implementation stage of environmental perception, autonomous navigation, and task control in orchard robots [134]. Unlike ordinary mobile platforms, operation-oriented orchard robots must not only reach target areas but also perform specific tasks, including harvesting, spraying, pollination, and weeding, within complex canopies, narrow rows, and unstable external environments. Operation outcomes are jointly affected by target-recognition accuracy, robot-localization error, actuator response speed, manipulator-control precision, and environmental disturbances. Precision operation should therefore not be interpreted as the design of an isolated actuator but as a closed-loop system encompassing perception and localization, motion control, operation mechanisms, and performance evaluation.

4.1. Perception, Localization, and Manipulator Control for Harvesting Robots

Harvesting is among the orchard-robot tasks with the most stringent requirements for perception, localization, and manipulator control. Fruits are often occluded by branches and leaves and distributed irregularly in three-dimensional space. The end effector must approach, grasp, detach, and avoid obstacles within a constrained workspace [135,136]. If three-dimensional fruit localization is inaccurate, stable harvesting cannot be achieved even with a highly precise manipulator. Conversely, if manipulator positioning error is large, the visually estimated target position will not coincide with the actual end-effector position [137].

For fruit localization, Bhattacharya et al. proposed a detection-segmentation-clustering framework for three-dimensional apple localization in complex orchards. Apples were first detected using a VLM, instance segmentation was then used to obtain fruit-pixel regions, and the pixels were projected into three-dimensional space. DBSCAN point-cloud clustering was subsequently applied to suppress depth noise [38]. The dataset contained 219 apples and covered indoor and outdoor scenes as well as real and simulated fruit. The method achieved a mean absolute localization error of 7.7 mm, a 74.3% improvement over the best baseline, and was deployed on a field harvesting robot. This study demonstrates that harvesting perception should not stop at two-dimensional bounding boxes but should output three-dimensional coordinates directly usable by the manipulator. However, the limited sample size means that generalization across cultivars, dense fruit occlusion, and seasonal variation requires further validation.

Regarding manipulator-control accuracy, Qiao et al. proposed a rigid-flexible coupled error-calibration method for a long-reach apple-harvesting manipulator. They attributed manipulator errors to manufacturing and assembly deviations, compliance deformation induced by self-weight and payload, and nonstructural errors such as backlash [138]. A rigid-flexible coupled pose-error model was established by integrating systematic error, self-weight compliance error, and payload compliance error. Iterative least squares and a residual neural network were then used for parameter identification and nonstructural-error estimation, respectively. In simulation, the identification accuracy of the model after removal of redundant parameters reached 93.478%. Physical experiments showed that, after compensation, the mean position error and position RMSE were 0.278 and 0.202 mm, respectively, while the mean orientation error and orientation RMSE were 0.230 and 0.178 degrees, respectively. Relative to the uncalibrated system, the mean position error and position RMSE decreased by 95.221% and 95.182%, and the mean orientation error and orientation RMSE decreased by 94.080% and 93.541%, respectively. These findings show that high-precision control of orchard-harvesting manipulators depends not only on kinematic models but also on compensation for flexible deformation and nonstructural errors introduced by lightweight structures. The method provides high-precision error

compensation for harvesting robots, although practical orchard harvesting still requires integration with visual servoing, obstacle-avoidance planning, and adaptive end effectors.

Overall, the core of precision harvesting lies in coordination between three-dimensional target localization and high-accuracy manipulator control. Future studies should evaluate the relationships among fruit-localization error, manipulator-compensation error, and harvesting success rather than assessing vision algorithms and mechanical systems separately.

4.2. Variable-Rate Application and Targeted Control for Spraying Robots

Spraying is one of the most widely implemented robotic tasks in orchard management. Its objectives are to reduce pesticide waste, improve the uniformity of canopy deposition, and decrease environmental contamination. Conventional continuous spraying generally uses fixed flow and air rates and readily causes overapplication, spray drift, and off-target deposition. Variable-rate spraying robots must instead adjust spraying parameters dynamically according to canopy morphology, target volume, distance, and operation zone [139,140].

Hui Liu et al. developed an intelligent multivariable spraying robot for orchards and nurseries. The system comprised a tracked mobile chassis, a navigation system, four independent spraying units, four centrifugal nozzles, four air-assisted units, a 75 L chemical tank, a three-dimensional LiDAR target-detection unit, and a central processing unit [40]. Three-dimensional LiDAR was used to acquire orchard point clouds, and an improved DBSCAN algorithm extracted target plants. A multivariable spraying model was then established from target position, shape, and volume to independently control spray flow rate, air volume, droplet diameter, and spraying direction. The oscillating-fan mechanism provided a maximum swing angle of 30 degrees, thereby increasing the spray range.

Field trials conducted with canopy diameters of 0.8–1.5 m, tree heights of 1.5–1.8 m, and inter-tree spacing of 2 m showed that conventional spraying consumed 180–220 g of liquid per nozzle, whereas multivariable spraying consumed only 30–35 g, corresponding to an average pesticide-saving rate of 83%. The study further reported that the coverage rate of multivariable spraying was lower than that of conventional spraying but higher than that of flow-only variable-rate spraying. At the same liquid application rate, variable air volume and adjustable droplet size improved penetration and reduced drift, resulting in more uniform coverage than flow-only variable spraying. The principal contribution of this study was the direct conversion of three-dimensional point-cloud target recognition into coordinated control of flow, air volume, droplet size, and spray direction, thereby establishing a relatively complete precision-spraying loop [141,142]. Remaining limitations include the sensitivity of canopy-volume estimation and liquid-deposition patterns to tree morphology, airflow, and sensor resolution. More accurate canopy modeling and deposition feedback are needed.

4.3. Flower Detection and Precision Operation for Pollination Robots

Pollination has strict timeliness and accuracy requirements. In kiwifruit, for example, flowers are distributed beneath the trellis, exhibit diverse orientations, and are frequently occluded by the canopy. Pollination performance is also affected by wind speed, wind direction, and spray volume [143]. Pollination robots must therefore simultaneously address flower recognition, spray localization, quantitative liquid control, and compensation for environmental disturbances [144].

Hao et al. developed a precision liquid-pollination robot for kiwifruit orchards. The system used a CLIP-assisted automatic-annotation method to construct a kiwifruit-flower detection dataset and trained a lightweight YOLO model for real-time recognition. A liquid-spray control system and a wind-compensation system were also designed to adjust spray offset and pollen dosage according to meteorological data [39]. The control architecture

comprised flower detection and manipulator control, wind-speed and wind-direction compensation, and programmable logic controller (PLC)-based quantitative liquid spraying. Wind-tunnel experiments were used to establish models of spray-center displacement and pollen compensation, reducing wind-induced pollen drift in open orchards. Figure 4 shows the demonstration of YOLO deployment and inference results on the platform.

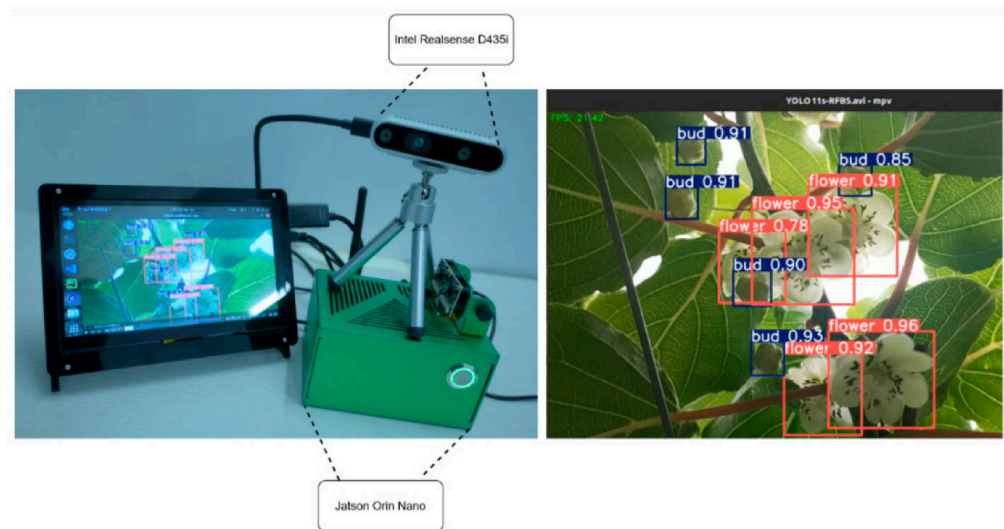


Figure 4. Demonstration of YOLO deployment and inference results on the platform [144].

Field trials conducted during 2024–2025 showed that the robot achieved commercial-standard fruit set, with rates exceeding 85%, pollen consumption of approximately 200 g/ha, and an operation speed of approximately 35–40 s/m². Under the assumption of 500,000 flowers per hectare, the study estimated that a single complete pollination pass would require approximately 110 h; one additional supplemental pass would increase the total operation time to approximately 220 h. These results demonstrate advantages in pollen conservation and pollination precision but also identify operation efficiency as a major constraint on large-scale application. The authors further noted that the current manipulator arrangement produced an ineffective workspace approximately 220 mm in diameter at the center of each operation zone, potentially leaving some flowers untreated. Raised beds around kiwifruit roots and fruiting branches near the main trunk also increased the difficulty of treating peripheral flowers. Overall, the study integrated flower detection, localized manipulator spraying, and wind compensation into an engineering-complete precision-pollination system, although manipulator coverage, mobile-operation efficiency, and fruit-quality evaluation require further improvement.

The application of multi-objective optimization in spraying systems remains in its infancy. Most existing studies aggregate multiple objectives into a single weighted target for resolution; while this approach simplifies the computational process, it risks discarding critical information along the Pareto frontier. Establishing systematic trade-off analyses among competing objectives—such as maximizing deposition, minimizing drift, and curtailing economic costs—constitutes a crucial trajectory for future exploration.

4.4. Intra-Row Navigation and Actuator Control for Weeding Robots

Intra-row orchard weeding is a representative near-tree operation. Conventional mowers cannot easily access the area immediately surrounding trunks, and residual weeds often require manual removal. Conversely, poorly controlled mechanical weeding mechanisms may damage trunks and roots [145,146]. Intra-row weeding robots must therefore coordinate tree-row tracking, trunk detection, lateral-mechanism control, and safe obstacle avoidance.

Sampurno et al. designed a single-tree-row-tracking robot for intra-row orchard weeding. The compact platform had a mass of 25 kg excluding payload, was powered by a 200 W portable battery, and used four-wheel independent drive with parallel front-wheel steering to operate beneath low canopies and in confined spaces [47]. A three-dimensional camera and a lightweight YOLO instance-segmentation model detected trunks in real time and generated a path with a lateral offset of 0.8 m from the tree row. LiDAR simultaneously triggered a laterally extending mowing mechanism, allowing the cutter to reach uncut weeds between trees. Figure 5 shows the working mechanism of a real-time, laser-based weed detection system.

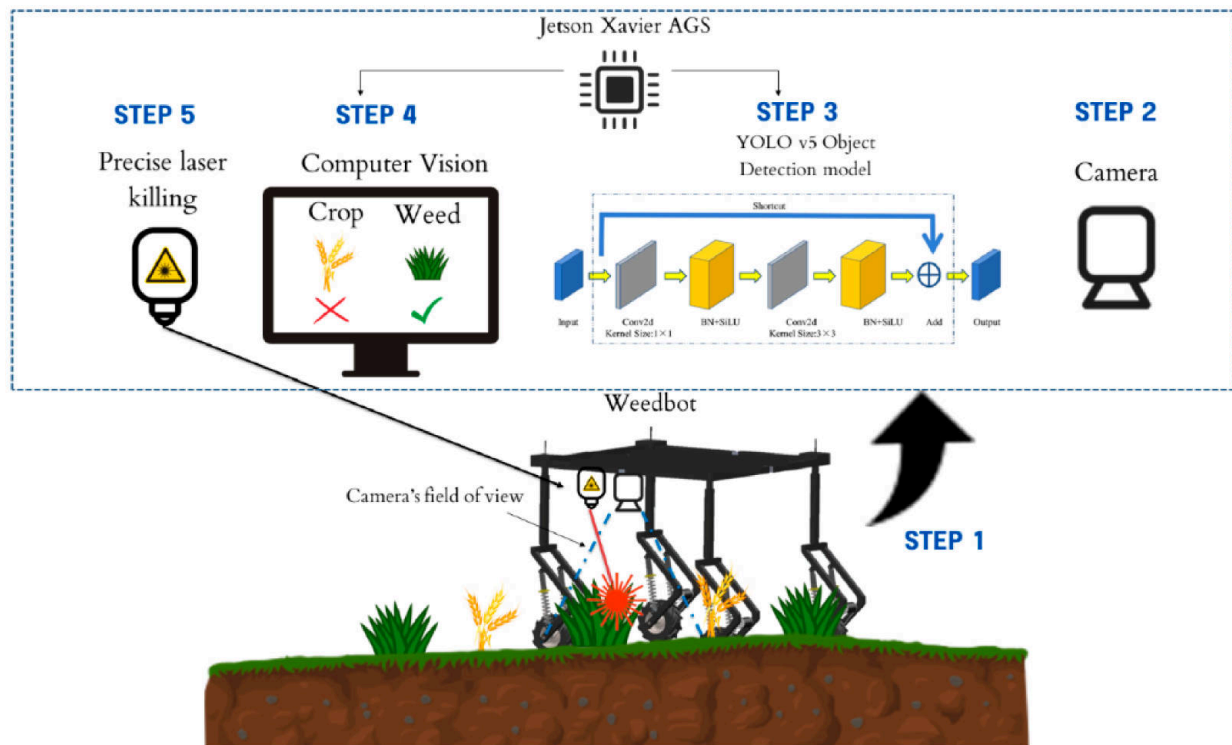


Figure 5. Functioning of a real-time, laser-based weed detection system [147].

In experiments with simulated irregular tree rows, the robot maintained relatively stable lateral tracking, with a mean lateral RMSE of 0.329 m and a mean heading RMSE of 4.45 degrees. The lateral weeding mechanism achieved an accuracy of 84.04% at 0.5 m/s, which decreased to 76.85% at 0.8 m/s. The authors attributed the reduced accuracy at higher speed to delays in real-time LiDAR point-cloud processing and actuator motion. These results show that the key challenge in intra-row weeding is not simply trunk or weed recognition but temporal matching among perception, control, and mechanical action. The main advantage of the study was the integration of visual navigation and LiDAR-triggered actuation on a single platform, providing a complete solution for intra-row weeding. However, lateral-tracking error remained relatively large, and orchards with young trees or narrow inter-tree spacing require more precise near-tree safety control.

The execution technologies used for four representative orchard tasks-harvesting, spraying, pollination, and intra-row weeding-differ substantially in technical emphasis and performance constraints. Their system configurations, experimental outcomes, and remaining limitations are summarized in Table 4 to clarify their technical characteristics and application value.

Table 4. Precision-execution technologies and performance of representative orchard robots.

Operation Task	Representative Study	System Configuration/ Key Technology	Experimental Result	Main Contribution	Remaining Limitation	Crop and Validation Context	Standardized Evidence Profile (Maturity; Field Scale; Robustness; Reproducibility; Computation; Applicability; NR = Not Reported)
Apple-harvesting localization	Bhattacharya et al.	VLM detection, instance segmentation, RGB-D, and DBSCAN clustering	Mean absolute localization error of 7.7 mm; 74.3% improvement over the best baseline	Provided three-dimensional fruit coordinates directly usable by a manipulator	Sample size and cross-scene generalization require further improvement	Apple; RGB-D dataset with 219 apples; indoor/outdoor and real/simulated samples; field harvesting-robot deployment. Quantitative occlusion level and validation duration: NR.	Field-deployed perception prototype; 219 apples in mixed real/simulated scenes; severe occlusion and cultivar transfer NR; data/code NR; real-time, hardware/FPS NR; direct harvesting coordinates.
Harvesting-manipulator calibration	Qiao et al.	Rigid-flexible coupled error model, iterative least squares, and residual neural network	Mean position error of 0.278 mm and mean orientation error of 0.230 degrees; position error reduced by 95.221%	Substantially improved localization accuracy of a long-reach harvesting manipulator	Requires joint evaluation with visual servoing and harvesting success	Long-reach apple-harvesting manipulator; simulation and physical calibration experiments. Orchard field setting, sample size, illumination, occlusion, operating speed, terrain, and long-term validation: NR.	Physically validated calibration subsystem; simulation plus manipulator tests, no orchard harvesting trial; environmental/visual-servo robustness NR; data/code NR; runtime/hardware NR; calibration-specific.
Variable-rate spraying	Hui Liu et al.	Three-dimensional LiDAR, improved DBSCAN, and a multivariable spraying model	30–35 g for multivariable spraying versus 180–220 g for conventional spraying; 83% pesticide saving	Coordinated control of flow rate, air volume, droplet diameter, and spraying direction	Canopy-volume estimation and deposition feedback require further improvement	Orchard and nursery field trials; canopy diameter 0.8–1.5 m, tree height 1.5–1.8 m, and 2 m inter-tree spacing. Crop/cultivar, operating speed, illumination, terrain, and validation duration: NR.	Field prototype; orchard/nursery trials with canopy geometry; cultivar/weather/terrain transfer NR; data/code/hardware NR; processing rate NR; variable-rate spraying under comparable canopies.
Kiwifruit pollination	Hao et al.	CLIP-assisted annotation, YOLO detection, manipulator spraying, and wind compensation	Fruit-set rate above 85%, pollen consumption of approximately 200 g/ha, and speed of 35–40 s/m ²	Integrated flower detection, wind compensation, and quantitative liquid spraying	Operation efficiency and manipulator coverage remain limited	Kiwifruit; open-orchard field trials during 2024–2025, with wind-tunnel calibration and wind compensation; operation speed 35–40 s/m ² . Dataset size, illumination/occlusion levels, terrain complexity, and continuous validation duration: NR.	Field prototype; orchard trials in 2024–2025 plus wind-tunnel calibration; wind robustness tested, lighting/occlusion/terrain NR; data/code/hardware NR; 35–40 s/m ² ; kiwifruit pollination.
Intra-row weeding	Sampurno et al.	Three-dimensional vision-based trunk detection, LiDAR triggering, and a lateral mowing mechanism	Accuracy of 84.04% at 0.5 m/s and 76.85% at 0.8 m/s	Coordinated navigation with a near-tree operation mechanism	Performance at high speed is strongly affected by point-cloud processing and actuator delay	Real-orchard intra-row weeding; operation evaluated at 0.5 and 0.8 m/s. Crop/cultivar, illumination, occlusion level, terrain complexity, sample size, and validation duration: NR.	Field prototype; real-orchard tests at 0.5 and 0.8 m/s; broader crop/terrain transfer NR; data/code NR; point-cloud/actuator delay limits speed; low-speed intra-row weeding.

4.5. Operation-Accuracy Calibration, Performance Evaluation, and System Integration

Operation accuracy in orchard robots is jointly determined by perceptual localization, navigation error, mechanical execution, and environmental disturbances and cannot be evaluated using a single algorithmic metric [147]. For harvesting robots, Qiao et al. established a rigid-flexible coupled pose-error model for a long-reach harvesting manipulator and incorporated manufacturing and assembly errors, self-weight deformation, payload-induced compliance deformation, and nonstructural errors into the compensation framework [138]. After compensation, the mean manipulator position error was 0.278 mm and the mean orientation error was 0.230 degrees; relative to the uncalibrated system, these errors decreased by 95.221% and 94.080%, respectively. These results indicate that lightweight manipulators used in orchards must account for structural flexibility and load variation. Otherwise, even accurate fruit localization may not enable the end effector to reach the target reliably.

For spraying and pollination robots, evaluation should place greater emphasis on agricultural outcomes. The multivariable spraying robot of Hui Liu et al. used three-dimensional LiDAR to extract target-plant information and controlled flow rate, air volume, droplet diameter, and spraying direction [40]. In field trials, conventional spraying consumed 180–220 g of liquid per nozzle, whereas multivariable spraying consumed 30–35 g, yielding an average pesticide-saving rate of 83%. The kiwifruit-pollination robot of Hao et al. integrated flower detection, wind compensation, and quantitative liquid spraying on a single platform and achieved a field fruit-set rate above 85% with pollen consumption of approximately 200 g/ha. These results demonstrate that precision-operation evaluation should extend beyond detection accuracy to include resource consumption and agronomic effectiveness [148].

Weeding robots further illustrate the influence of system response on operation accuracy [149]. The intra-row weeding robot of Sampurno et al. achieved 84.04% lateral-weeding accuracy at 0.5 m/s, but accuracy decreased to 76.85% when speed increased to 0.8 m/s, primarily because of delays in point-cloud processing and mechanism actuation. Evaluation of intra-row operation should therefore consider not only path tracking but also actuator response, the risk of missed treatment, and trunk safety.

Collectively, existing studies suggest that orchard-robot performance evaluation should include module-level, operation-level, and production-level metrics. Module-level metrics include detection accuracy, localization error, and control latency. Operation-level metrics include harvesting success, pesticide savings, fruit-set rate, and weeding accuracy. Production-level metrics include operation time per unit area, resource consumption, and long-term stability [150]. An integrated assessment is more likely to provide a comprehensive evaluation field applicability.

5. Current Challenges and Future Prospects

5.1. Robustness Challenges in Perception, Navigation, and Operation Under Complex Orchard Conditions

Existing studies demonstrate substantial progress in target recognition, localization and navigation, and operation execution; nevertheless, complex orchard environments remain the principal barrier to stable deployment. Fruits, flowers, branches, trunks, and weeds are frequently affected by occlusion, overlap, and illumination variation, resulting in inconsistent perception across tree forms, cultivars, and seasons. Bhattacharya et al. integrated a VLM, instance segmentation, and DBSCAN clustering for three-dimensional apple localization and achieved a mean absolute error of 7.7 mm, a 74.3% improvement over the best baseline. This result confirms that multimodal perception can improve localization accuracy in complex environments. However, the dataset contained only 219 apples, and generalization across orchards, cultivars, and severe occlusion remains to be validated.

For navigation, canopy occlusion and terrain undulation reduce GNSS reliability and compel robots to rely on onboard vision, LiDAR, IMUs, and odometry. The jujube-orchard spraying robot fused LiDAR SLAM, an IMU, EKF, and AMCL and achieved a mean X-axis deviation of 3.41 cm along a 15 m straight path and 1.8 cm along an S-shaped path, demonstrating the effectiveness of multi-sensor fusion for cruising in densely planted orchards. However, long-term navigation stability has not been adequately validated in more challenging environments involving sloped terrain, muddy ground, and dynamic obstacles.

Operation execution is likewise affected by environmental disturbances. The kiwifruit-pollination robot combined flower detection, wind compensation, and quantitative liquid spraying to achieve a field fruit-set rate above 85% with pollen consumption of approximately 200 g/ha. The study nevertheless reported an ineffective manipulator workspace approximately 220 mm in diameter, which could leave some flowers untreated, and the operation time remained excessive for large orchards. The weeding robot achieved 84.04% lateral-weeding accuracy at 0.5 m/s, but accuracy decreased to 76.85% at 0.8 m/s, demonstrating that perception-processing delay and actuator response directly affect operation quality. Future research should therefore strengthen closed-loop coordination among perception, navigation, and actuation to improve continuous operation in real orchards.

5.2. Bottlenecks in Multi-Robot Collaborative Scheduling and Intelligent Decision-Making

As orchard size increases, a single robot cannot satisfy the time-critical requirements of many operations, making multi-robot and robot-UAV collaboration increasingly important. However, most existing studies focus on static task allocation and simulation-based optimization and do not adequately consider dynamic task changes, equipment failure, battery replenishment, communication latency, or weather.

For multi-robot scheduling, Chen et al. investigated multi-trip task scheduling for harvesting robots in smart orchards. They considered the relationships among robot self-weight, real-time payload, and energy consumption and proposed an adaptive experience-based discrete genetic algorithm. Tests on 18 newly generated orchard instances and 24 existing benchmark instances showed that the algorithm outperformed six advanced alternatives. The study realistically represented repeated unloading and re-entry of harvesting robots, but validation remained predominantly instance- and simulation-based and did not fully account for path blockage, robot failure, or temporary task changes in real orchards.

For robot-UAV collaboration, Wang et al. studied task allocation for pollination robots and spraying UAVs in multi-orchard scenarios. They formulated a multi-orchard robot-drone task-allocation (MORDTA) model that minimized the maximum completion time across all orchards and proposed an improved evolutionary algorithm incorporating problem-specific knowledge. A total of 500 instances were generated, of which 125 were used for parameter calibration and 375 for testing; the proposed method outperformed several comparison algorithms. The model incorporated the precedence constraint that ground pollination must be completed before aerial spraying, providing strong scenario specificity. However, it assumed constant robot and UAV speeds, normal equipment operation, and negligible equipment-transport time, leaving a gap between the model and practical deployment.

For intelligent decision-making, LLMs and VLMs offer a new route to natural-language task planning. Salman et al. proposed the AGRI-BT robot framework, which used LoRA-fine-tuned CodeLlama-7B to generate XML behavior trees and CLIP for visual-semantic judgment. The system achieved 100% XML structural validity, 87.5% semantic correctness, 82.0% task alignment, and an overall weighted validation score of 91.4%. However, alignment on complex multistage tasks remained lower than structural validity, indicating that foundation models in agricultural robotics must be combined with rule constraints, be-

havior trees, semantic maps, and safety-control mechanisms rather than directly replacing verifiable robot-control systems. Multi-robot scheduling, ground-air collaboration, and foundation-model-enabled decision-making constitute three major directions of collaborative intelligence for orchard robots. Representative studies are compared in Table 5 to summarize differences in problem formulation, solution methods, validation settings, and current bottlenecks.

5.3. Data Benchmarks, System Integration, and Future Research Directions

Orchard-robot research currently lacks unified datasets, evaluation metrics, and long-term field-testing benchmarks. Different studies use detection accuracy, localization error, path deviation, pesticide-saving rate, fruit-set rate, weeding accuracy, and completion time. Although these metrics characterize individual systems, they make cross-system comparison difficult. Recent work has begun to emphasize data and benchmark development. Chen et al. constructed 18 new orchard scheduling instances and combined them with 24 existing benchmarks to evaluate multi-trip harvesting schedules. Orchard size ranged from 20 m × 20 m to 70 m × 70 m, the number of trees ranged from 100 to 1225, and fruit-maturity ratios were set to 0.4, 0.6, and 0.8, providing a basis for algorithm comparison. AGRI-BT robot established 1187 agricultural-task descriptions paired with expert-validated behavior trees, and Hao et al. released code and data resources for their kiwifruit-pollination robot. These efforts improve research reproducibility.

Robotic pruning, advanced manipulation, and Vision–Language–Action systems represent important emerging directions for orchard robotics. Compared with harvesting, spraying, and weeding, robotic pruning requires a more detailed understanding of tree architecture and agronomic objectives. A pruning robot must reconstruct branch topology, identify appropriate pruning points, evaluate branch accessibility, generate collision-free cutting trajectories, and accurately control the cutting tool without damaging retained branches or nearby fruits. Variations in branch diameter, orientation, stiffness, and occlusion further increase the difficulty of autonomous pruning. Future studies should therefore integrate three-dimensional structural perception, agronomic decision rules, manipulator planning, and cutting-quality evaluation within a unified closed-loop system.

Advanced manipulation is also essential for improving the adaptability of orchard robots. Compliant control, force and tactile sensing, active perception, dexterous end effectors, and coordinated multi-arm systems may enable robots to interact more safely and effectively with fragile, deformable, and irregularly distributed biological targets. In particular, the combination of visual feedback and contact information could help compensate for localization errors and unexpected target motion during harvesting, pruning, and flower manipulation. However, the increased mechanical and computational complexity of such systems must be balanced against real-time performance, maintenance requirements, and field reliability.

Beyond the use of LLMs and VLMs for semantic understanding and task planning, emerging Vision–Language–Action systems provide a potential framework for mapping visual observations and natural-language instructions directly to executable robotic actions. Such systems may support more flexible task specification, cross-task knowledge transfer, and adaptive decision-making in changing orchard environments. Nevertheless, orchard-oriented VLA research remains at an early stage. Major challenges include the scarcity of orchard-specific action datasets, domain shifts across cultivars, seasons, and orchard structures, high computational requirements, limited interpretability of generated actions, and the need to guarantee physical safety. Closed-loop field validation and the integration of explicit geometric, agronomic, and safety constraints will therefore be necessary before VLA systems can be reliably deployed in practical orchard operations.

Table 5. Comparison of representative studies on collaborative scheduling and intelligent decision-making for orchard robots.

Research Direction	Study	Problem Type	Method	Experimental Setting/Data	Main Result	Development Bottleneck	Validation Context and Key Assumptions	Standardized Evidence Profile (Maturity; Field scale; Robustness; Reproducibility; Computation; Applicability; NR = Not Reported)
Multi-trip harvesting scheduling	Chen et al.	Multi-robot, multi-trip harvesting-task scheduling	adaptive experience-based discrete genetic algorithm	18 new orchard instances and 24 benchmark instances	Outperformed six advanced algorithms	Primarily simulation based; robot failure and dynamic tasks were not fully considered	Simulation/benchmark evaluation; 18 new orchard instances plus 24 existing benchmarks; orchard size 20 m × 20 m to 70 m × 70 m; 100–1225 trees; maturity ratios 0.4, 0.6, and 0.8. No physical field-duration validation.	Simulation-stage method; 42 generated/benchmark instances; orchard size, tree count, and maturity varied, but failures/dynamic tasks excluded; code/data NR; runtime/scalability NR; field validation required.
Joint task-path optimization	Li et al.	Agricultural-robot task allocation and path planning	Improved eight-direction A*, Bezier curves, and a multi-objective discrete artificial bee colony algorithm	Simulated agricultural task scenarios	Reduced operation cost and improved task balance	Obstacles and dynamic interactions in real environments require validation	Simulated agricultural task scenarios. Crop/orchard architecture, sensor configuration, travel speed, terrain complexity, environmental variability, sample size, and field-validation duration: NR.	Simulation-stage method; simulated task scenarios only; real obstacles, dynamics, terrain, and weather unvalidated; detailed instances/code NR; runtime/scalability NR; field validation required.
Robot-UAV collaboration	Wang et al.	Task allocation for pollination robots and spraying UAVs across multiple orchards	Improved evolutionary algorithm incorporating problem-specific knowledge	500 instances: 125 for parameter calibration and 375 for testing	Outperformed multiple comparison algorithms	Assumed constant speed and normal equipment operation and did not adequately represent real constraints	Generated multi-orchard robot–UAV instances; 500 total (125 for parameter calibration and 375 for testing). Constant platform speeds, normal equipment operation, and negligible transport time were assumed; no physical field validation.	Simulation-stage scheduler; 500 generated instances; constant speed, normal operation, and negligible transport assumed; code/data NR; runtime/scalability NR; applicable only under stated assumptions.
LLM/VLM-based intelligent decision-making	Salman et al.	Natural-language task planning and visual decision-making for agricultural robots	LoRA-fine-tuned CodeLlama-7B, CLIP, and behavior trees	1187 agricultural-task descriptions paired with expert-validated behavior trees	XML validity 100%, semantic correctness 87.5%, and task alignment 82.0%	Alignment remains insufficient for complex multistage tasks	Greenhouse-oriented AGRI-BT framework; 1187 natural-language task/behavior-tree pairs; ROS 2 simulation and natural-versus-Gazebo image evaluation. No orchard field validation; deployment duration and physical-platform speed: NR.	Simulation-stage greenhouse framework; 1187 task/tree pairs plus ROS 2 tests; natural-to-Gazebo domain gap observed; public code/data/model NR; onboard runtime/hardware NR; orchard transfer unvalidated.

Digital twins, realistic simulation environments, and sim-to-real learning are becoming important tools for accelerating the development of orchard robots. A digital twin can provide a virtual representation of the orchard, robotic platform, sensors, actuators, and operational processes and may be updated using data collected from field robots and environmental-monitoring systems. Such a framework can support robot design, mission planning, prediction of operational performance, maintenance analysis, and evaluation of alternative control or scheduling strategies before physical deployment. For large orchards and multi-robot systems, digital twins may also facilitate fleet-level monitoring, resource allocation, and the analysis of interactions among robots, crops, terrain, and environmental conditions.

Realistic simulation environments provide a complementary approach for developing and testing robotic systems under repeatable and controllable conditions. Virtual orchards can represent variations in tree architecture, canopy density, fruit and flower distribution, illumination, weather, terrain slope, soil conditions, occlusion, and sensor noise. They can therefore support large-scale generation of training data and systematic evaluation of perception, navigation, manipulation, and collaborative-planning algorithms without the cost, seasonal restrictions, or safety risks of continuous field experimentation. Simulation is particularly valuable for rare or hazardous scenarios, such as localization failure, severe wheel slip, manipulator collision, communication loss, and interactions between multiple robots.

However, models trained or validated in simulation often experience reduced performance after deployment because virtual scenes cannot fully reproduce the visual appearance, physical properties, and dynamic variability of real orchards. Sim-to-real learning methods, including domain randomization, synthetic-data generation, transfer learning, system identification, and progressive adaptation using limited field data, may help reduce this gap. For manipulation tasks, accurate simulation of branch flexibility, fruit detachment, contact forces, cutting processes, and end-effector interactions remains especially challenging. Similarly, terrain simulation must account for soil deformation, vegetation cover, wheel slip, sinkage, and changing moisture conditions. Future research should therefore integrate digital twins and simulation-based learning with hardware-in-the-loop experiments, real sensor data, and staged field trials. Simulation should be treated as a tool for accelerating development and risk reduction rather than as a substitute for long-term, multi-season field validation.

Future orchard-robot research should prioritize three directions. First, multimodal datasets should be established across crops, seasons, and orchards, covering images, point clouds, semantic maps, robot trajectories, and operation outcomes. Second, a unified system-level evaluation framework should integrate perception accuracy, navigation stability, execution success, resource consumption, and agronomic effectiveness. Third, long-term field validation of complete robotic systems should be strengthened, with particular attention to stability under dust, post-rain ground conditions, canopy occlusion, dense fruit clusters, and continuous operation. Orchard robots can progress from prototype validation toward scalable application only when algorithmic performance, operation quality, and production benefits are evaluated together. Table 6 summarizes current bottlenecks and potential research breakthroughs across six dimensions: environmental perception, autonomous navigation, precision operation, multi-robot collaboration, intelligent decision-making, and evaluation benchmarks.

The quantitative results summarized in Tables 2–5 should be interpreted as study-specific evidence rather than as directly comparable benchmarks. The reviewed studies differ substantially in crop species and cultivars, orchard architecture, sensing configuration, dataset size, laboratory or field setting, illumination and occlusion conditions, terrain complexity, robot travel speed, validation duration, and metric definitions.

Table 6. Current challenges and future research directions for orchard robots.

Research Component	Current Challenge	Future Research Direction
Environmental perception	Occlusion, overlap, illumination variation, and insufficient cross-cultivar generalization	Establish multi-season, multi-orchard, and multi-crop datasets; develop multimodal perception and foundation-model adaptation methods
Autonomous navigation	GNSS constraints, uneven terrain, irregular tree rows, and difficulty in long-term mapping	Fuse vision, LiDAR, IMUs, and semantic maps; improve long-term localization and dynamic map-updating capabilities
Precision operation	Manipulator error, spray drift, low pollination efficiency, and delayed weeding-actuator response	Establish closed loops linking perception, control, and execution; develop compliant control, error compensation, and operation-feedback mechanisms
Multi-robot collaboration	Predominantly static scheduling and insufficient consideration of real dynamic constraints	Introduce online rescheduling, energy management, communication constraints, and fault-recovery mechanisms
Intelligent decision-making	Insufficient task alignment and safety verifiability of LLMs/VLMs	Integrate foundation models with behavior trees, semantic maps, and rule constraints to improve interpretability and reliability
Evaluation benchmarks	Fragmented metrics and a lack of unified field-testing standards	Construct system-level evaluation frameworks covering perception, navigation, execution, operation outcomes, and production benefits

5.4. Technology Readiness and Barriers to Commercial Deployment

The transition of orchard robots from research prototypes to commercially deployable systems depends not only on algorithmic accuracy and operation performance but also on technological maturity, economic feasibility, human factors, regulatory compliance, and organizational capacity. The studies reviewed in this article exhibit substantially different levels of maturity. To avoid assigning unsupported numerical technology-readiness levels, the available evidence can be qualitatively divided into four stages: algorithm- or simulation-level validation, controlled-environment prototype validation, single-orchard field-prototype validation, and multi-orchard or long-term pre-commercial validation.

According to this framework, many fruit-, flower-, branch-, and trunk-perception methods have reached the controlled-environment or single-orchard prototype stage. Several LiDAR-, vision-, and multisensor-based navigation systems have also been tested using integrated mobile platforms under real orchard conditions. Nevertheless, their robustness across cultivars, seasons, orchard architectures, terrain conditions, and prolonged operating periods remains insufficiently demonstrated. Harvesting, pollination, spraying, and intra-row weeding systems have progressed from isolated algorithm validation toward integrated field prototypes, but their commercial readiness is still constrained by cycle time, actuator durability, operation coverage, maintenance requirements, and the limited duration and scale of field validation. Multi-robot coordination, robot-UAV scheduling, and foundation-model-assisted decision-making generally remain at earlier maturity levels because much of the current evidence is derived from optimization benchmarks, simulations, or constrained demonstrations rather than long-term closed-loop operation in real orchards.

Economic feasibility is another major barrier to adoption. The total cost of an orchard robot extends beyond its initial purchase price and includes sensors, manipulators, batteries, charging or refilling facilities, software, routine calibration, maintenance, component replacement, technical support, and operator training. High-value sensors and precision actuators may improve performance but also increase acquisition and repair costs. Furthermore, orchard operations are strongly seasonal. A robot designed for only one short-duration task may remain unused during much of the year, reducing equipment utilization and delaying the recovery of capital investment. Consequently, return

on investment depends on orchard area, crop value, labor availability, annual operating duration, operation speed, reliability, ownership model, and the number of tasks that can be performed by the same platform. These constraints are especially important for small- and medium-sized producers, who may have limited capital, technical personnel, repair facilities, and access to specialized after-sales services.

Improving commercial feasibility will therefore require not only higher technical performance but also modular and maintainable system design. Interchangeable operation modules may allow one mobile platform to support inspection, spraying, weeding, transportation, or other tasks at different times of the production cycle, thereby increasing seasonal utilization. Simplified calibration procedures, standardized interfaces, accessible replacement parts, remote diagnostics, and local maintenance support may also reduce downtime and lifecycle cost. Shared ownership, leasing, cooperative operation, and robot-as-a-service models may provide alternative adoption pathways when individual farms cannot justify purchasing and maintaining complete robotic systems. However, the economic effectiveness of these models requires farm-level and region-specific validation rather than inference from algorithmic performance alone.

Human factors and occupational safety must also be considered during system development. Orchard robots may operate near workers, trees, irrigation equipment, vehicles, and animals, while spraying robots additionally involve potential exposure to agricultural chemicals. Practical systems should therefore provide reliable obstacle and human detection, emergency stopping, manual override, safe-state transitions, clear status indicators, and procedures for recovery after sensor, actuator, communication, or localization failure. Operator interfaces should be understandable to users without specialist robotics training, and routine activities such as battery replacement, refilling, cleaning, calibration, and fault diagnosis should be designed to minimize physical and cognitive burden. Field validation should evaluate not only robot performance but also operator workload, required training time, intervention frequency, near-miss events, and the safety of maintenance procedures.

Compatibility with existing orchard infrastructure constitutes an additional deployment constraint. Row spacing, canopy form, turning space, slope, ground condition, irrigation pipelines, support structures, wireless coverage, and access to electricity or refilling stations vary substantially among orchards. A robot that performs well in a modern high-density orchard may be unsuitable for older orchards with irregular rows, narrow passages, steep terrain, or heterogeneous tree architecture. Commercial systems must therefore either adapt to existing orchards or clearly specify the infrastructural conditions required for deployment. Excessive dependence on orchard reconstruction would substantially increase adoption costs, particularly for small producers.

Regulatory and organizational issues may further delay commercialization. Depending on the platform and operation, orchard robots may be subject to requirements concerning agricultural-machinery safety, autonomous vehicle operation, pesticide application, UAV flight, electromagnetic compatibility, data protection, and occupational health. Liability must also be clarified when damage results from perception errors, software failure, communication loss, incorrect operator configuration, or interactions between multiple autonomous machines. Because these requirements vary across jurisdictions, commercialization requires cooperation among robot developers, orchard managers, machinery manufacturers, service providers, insurers, and regulatory authorities. Traceable operation logs, defined human-supervision responsibilities, risk-assessment procedures, and verifiable fail-safe mechanisms will be important for certification and accountability.

Overall, current orchard-robot research is advancing more rapidly in algorithmic capability than in lifecycle cost assessment, maintainability, human-centered design, regulatory preparation, and organizational integration. Future field studies should therefore

report not only accuracy and operation outcomes but also energy consumption, intervention frequency, downtime, maintenance workload, component failure, operator-training requirements, seasonal utilization, safety incidents, and economic performance. Long-term and multi-orchard studies incorporating these indicators are necessary to determine whether a technically successful prototype can become a safe, affordable, and operationally sustainable commercial system.

6. Conclusions

This review examined the development of smart orchard robots through an integrated technological chain comprising environmental perception, autonomous navigation, precision operation, and collaborative decision-making. Its principal contribution is the establishment of an engineering-oriented framework that connects sensing and semantic understanding with robot mobility, task execution, and system-level coordination. Rather than treating harvesting, spraying, pollination, weeding, and multi-robot operation as isolated applications, the review demonstrates that their performance depends on the coordinated operation of perception, localization, planning, actuation, and evaluation components.

The reviewed evidence indicates that different technological areas have reached substantially different levels of maturity. Selected fruit-, flower-, trunk-, and canopy-perception methods have been integrated with robotic platforms, while LiDAR-, vision-, and multisensor-based navigation systems have achieved credible, although mostly site-specific, field validation in GNSS-constrained orchards. Semantic mapping has also been demonstrated in greenhouse, apple-orchard, and litchi-orchard environments. Among operation technologies, variable-rate spraying and wind-compensated pollination have produced measurable field outcomes, and several harvesting and intra-row weeding systems have progressed to integrated prototype demonstrations. Nevertheless, most of these results were obtained using limited crops, orchard structures, seasons, operating speeds, or validation periods. They should therefore be interpreted as evidence of task-specific feasibility rather than proof of broad commercial readiness.

By contrast, heterogeneous multi-robot coordination, robot-UAV scheduling, large-scale collaborative operation, and LLM- or VLM-assisted task planning remain predominantly exploratory. Much of the supporting evidence is derived from optimization benchmarks, simulation, constrained datasets, or controlled demonstrations. These approaches show potential for improving task allocation, semantic reasoning, and system flexibility, but their reliability under communication loss, dynamic obstacles, equipment failure, environmental change, and long-term closed-loop operation has not yet been sufficiently established. Foundation models should therefore be regarded as high-level enabling tools rather than verified replacements for deterministic and safety-critical robotic control.

Based on the reviewed evidence, three priorities are particularly important. First, cross-orchard, cross-cultivar, and cross-season datasets and standardized evaluation protocols are required to support reproducible comparison and assess environmental generalization. Second, future systems should prioritize long-term closed-loop integration of perception, navigation, and actuation, including uncertainty estimation, real-time edge deployment, failure detection, safe-state transition, and human intervention mechanisms. Third, collaborative and foundation-model-assisted systems must progress from simulation and constrained demonstrations to physical validation involving heterogeneous platforms, realistic communication and energy constraints, dynamic task reassignment, and sufficiently long operation in real orchards.

Overall, the transition from successful prototypes to dependable orchard-robot systems will require a shift from isolated algorithm optimization toward system-level robustness, standardised evaluation, operational safety, maintainability, and sustained field validation.

Progress in these areas will determine whether smart orchard robots can evolve from crop- and site-specific demonstrations into adaptable, safe, and practically deployable agricultural systems.

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Abbreviations

The following abbreviations are used in this manuscript:

AMCL	Adaptive Monte Carlo Localization
CLIP	Contrastive Language–Image Pre-training
CSF	Cloth Simulation Filtering
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DWA	Dynamic Window Approach
EKF	Extended Kalman Filter
FPS	Frames Per Second
GNSS	Global Navigation Satellite System
IMU	Inertial Measurement Unit
IoU	Intersection over Union
LiDAR	Light Detection and Ranging
LIO-SAM	LiDAR–Inertial Odometry via Smoothing and Mapping
LLM	Large Language Model
LoRA	Low-Rank Adaptation
mAP	Mean Average Precision
mIoU	Mean Intersection over Union
MPA	Mean Pixel Accuracy
PLC	Programmable Logic Controller
RGB	Red–Green–Blue
RGB-D	Red–Green–Blue–Depth
RMSE	Root Mean Square Error
ROS	Robot Operating System
RRT	Rapidly Exploring Random Tree
SLAM	Simultaneous Localization and Mapping
TEB	Timed Elastic Band
UAV	Unmanned Aerial Vehicle
VFH	Viewpoint Feature Histogram
VLM	Vision–Language Model
YOLO	You Only Look Once

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